

A Comprehensive Review of Machine Learning, Deep Learning, and Explainable AI in Healthcare: Medical Diagnosis, Interpretability, and Clinical Challenges

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Abstract—Artificial intelligence (AI) has become an important tool for medical diagnosis, risk prediction, and clinical decision support. This review examines machine learning, deep learning, hybrid approaches, and Explainable AI (XAI), with emphasis on medical diagnosis, model interpretability, and clinical challenges. Studies published between 2013 and 2026 were identified through IEEE Xplore, PubMed/MEDLINE, Scopus, Web of Science, the ACM Digital Library, and Google Scholar. Following screening, 88 studies were included in the analysis. The reviewed work spans cardiology, neurology, oncology, dermatology, ophthalmology, and medical imaging, and includes traditional machine learning methods, convolutional and recurrent neural networks, vision transformers, and ensemble models. Several studies reported accuracies above 95% for selected diagnostic tasks, although results varied across datasets and validation settings. The review also considers SHAP, LIME, and Grad-CAM as methods for improving the interpretability of model predictions. In addition, it discusses the use of AI in personalized treatment and the main barriers to clinical adoption, including data privacy, algorithmic bias, limited external validation, regulatory concerns, workflow integration, and infrastructure constraints. Overall, the literature shows strong potential for AI-assisted healthcare, but broader clinical use will require more rigorous validation, transparent reporting, and careful evaluation in real-world settings.

Index Terms—Artificial Intelligence, Explainable AI (XAI), Medical Diagnostics, Personalized Treatment, Machine Learning, Deep Learning, Clinical Implementation, Medical Imaging Analysis

I. INTRODUCTION

Healthcare increasingly relies on large and heterogeneous sources of data, including medical images, physiological signals, genomic information, and clinical records. Extracting clinically useful patterns from these data has driven the adoption of machine learning (ML) and deep learning (DL) methods across a wide range of diagnostic and predictive tasks. These methods have been applied to cardiovascular risk prediction, neurological disease detection, cancer diagnosis, and medical image interpretation [1, 2, 3]. XAI has also become important because, in clinical settings, predictive performance alone may not be sufficient when the basis of a model's decision cannot be understood.

Despite promising results, moving AI models from experimental studies into clinical practice remains difficult. Per-

formance reported on a development or benchmark dataset may not hold across hospitals, patient populations, or clinical settings. Differences in data quality, disease prevalence, population characteristics, class distribution, and validation procedures can affect reported results. Other concerns include algorithmic bias, patient privacy, limited external validation, model interpretability, regulatory requirements, and integration with existing clinical workflows. These issues are especially important in healthcare, where an inaccurate or poorly understood prediction may influence patient management. Evaluating healthcare AI therefore requires attention not only to predictive performance, but also to the reliability, interpretability, and clinical applicability of the resulting models.

The main contribution of this review is to bring together evidence on diagnostic performance, model interpretability, and the clinical challenges that affect the adoption of AI in healthcare. We examine how ML, DL, and hybrid approaches have been used for diagnosis and prediction across cardiology, neurology, oncology, dermatology, ophthalmology, medical imaging, and other healthcare applications. We summarize the model families, data modalities, and performance measures reported in the literature and examine how SHAP, LIME, Grad-CAM, and related XAI methods are used to interpret model predictions. We also discuss factors that may limit clinical adoption, including external validation, bias, privacy, workflow integration, and regulatory requirements. By considering these issues together, the review provides a broader assessment of healthcare AI than one based on reported diagnostic accuracy alone. Personalized treatment and clinical decision support are discussed as related applications.

The remainder of the paper is organized as follows. Section II describes the review methodology, Section ?? provides the relevant background, and Section III reviews the main AI technologies considered in the literature. The subsequent sections examine applications across healthcare domains and XAI methods, followed by personalized treatment and clinical decision support in Section VII and future directions in Section VIII. Section IX concludes the review.

II. METHODOLOGY

This review adopts a comprehensive literature review approach to synthesize the current state of ML, DL and XAI research across major healthcare domains. Given the breadth of clinical applications and model architectures covered, a narrative synthesis was chosen over a formal systematic review protocol to allow for a wider, more integrative discussion of trends, techniques, and challenges across the field.

A. Literature Search Strategy

Literature was identified through searches of **IEEE Xplore**, **PubMed/MEDLINE**, **Scopus**, **Web of Science**, **ACM Digital Library** and **Google Scholar**. Search terms combined keywords related to machine learning, deep learning, explainable AI, healthcare, medical diagnosis, and predictive modeling, joined using Boolean operators, e.g., “machine learning” AND “healthcare” AND “diagnosis” OR “prediction”; “explainable AI” AND “SHAP” OR “LIME” OR “Grad-CAM” AND “clinical”. Only studies published in English between 2013 and 2026 were considered.

B. Inclusion and Exclusion Criteria

Studies were included if they: (1) applied a machine learning, deep learning, or hybrid AI model to a healthcare-related diagnostic, predictive, or clinical decision-support task; (2) reported quantitative performance metrics e.g., accuracy, sensitivity, specificity, AUC, F1-score; (3) were published in a peer-reviewed journal or conference proceedings and (4) were accessible in full text. Studies were excluded if they: (1) were not available in English or in full text; (2) were purely theoretical or opinion-based without reported empirical results; (3) fell outside the healthcare domain; or (4) were preprints without peer review, unless no peer-reviewed alternative on the same topic was available. Based on these criteria, 88 studies were selected for inclusion in the qualitative and comparative analysis presented in this review.

C. Data Extraction and Synthesis

For each included study, information was extracted on the healthcare domain addressed, the AI/ML model architecture used, key input features or data modalities, the dataset and its size, reported performance metrics, and the use of any explainability methods. Extracted data were synthesized narratively and organized by clinical domain (e.g., cardiovascular, neurological, oncological) and methodological theme (e.g., traditional ML, DL, hybrid/ensemble methods, XAI techniques), following established narrative synthesis practices for literature reviews that span heterogeneous study designs and outcome measures. This synthesis is reflected in the domain-specific subsections and summary tables (Tables III–V) presented later in this paper.

D. Research Questions

The review was guided by the following research questions:

- **RQ1:** Which ML, DL, and hybrid models are used for diagnosis and prediction across major healthcare domains?

- **RQ2:** How are the performance and validation of these models reported across different clinical tasks and data modalities?
- **RQ3:** Which XAI methods are used to interpret model predictions, and how are they applied across different models and healthcare applications?
- **RQ4:** What technical, ethical, regulatory, and implementation challenges affect the clinical adoption of AI-based healthcare systems?

III. AI TECHNOLOGIES IN HEALTHCARE

Healthcare AI draws on a range of computational approaches suited to different types of clinical data and prediction tasks. ML methods are commonly used with structured clinical data, whereas DL architectures are widely applied to medical images, physiological signals, and other high-dimensional data [4, 5]. Hybrid and ensemble approaches combine complementary models or data sources to address more complex prediction tasks. As illustrated in Fig. 1, AI is applied across areas such as early disease detection, personalized treatment, clinical decision support, remote healthcare, and patient monitoring, with data and feedback from these applications supporting an iterative process of system improvement. This section reviews the principal ML, DL, hybrid, and ensemble methods encountered in these healthcare applications, followed by the performance measures used to evaluate them.

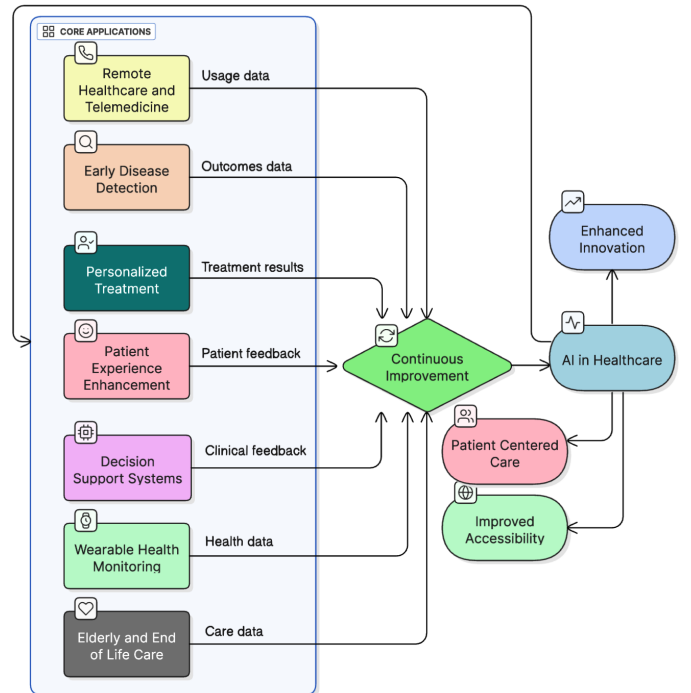


Fig. 1: AI-enabled healthcare applications and the role of data and feedback in continuous system improvement.

A. Machine Learning Algorithms for Healthcare

Traditional algorithms continue to play an important role in healthcare applications, particularly for structured clinical data, electronic health records, and predictive modeling tasks where interpretability is crucial. These algorithms offer computational efficiency, transparency, and efficient performance across different medical domains.

Logistic Regression & Support Vector Machines: These models are widely used in medical decision-making due to their complementary strengths. Logistic Regression [6] is commonly employed for clinical risk prediction and disease diagnosis because of its simplicity, interpretability, and ability to estimate the probability of an event, such as predicting the likelihood of heart disease from patient clinical parameters. The model uses the logistic (sigmoid) function to transform a linear combination of input features into a probability between 0 and 1. The Logistic Regression function is expressed as:

$$P(Y = 1|X) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_n X_n)}} \quad (1)$$

where:

- 1) $P(Y = 1|X)$ is the probability that the input sample belongs to the positive class, e.g., presence of heart disease.
- 2) Y is the target variable representing the class label.
- 3) $X = (X_1, X_2, \dots, X_n)$ denotes the vector of input features such as age, cholesterol level, blood pressure, maximum heart rate, and ECG measurements.
- 4) β_0 is the intercept (bias) term.
- 5) $\beta_1, \beta_2, \dots, \beta_n$ are the coefficients (weights) corresponding to each input feature, indicating their contribution to the prediction.
- 6) n represents the total number of input features.
- 7) e is Euler's number (approximately 2.71828), which forms the basis of the natural logarithm.

A positive coefficient increases the probability of the positive class, whereas a negative coefficient decreases it. The magnitude of each coefficient reflects the strength of the relationship between the corresponding feature and the predicted outcome. Support Vector Machines (SVMs) [7] are powerful supervised learning algorithms designed for binary classification. They are particularly effective for high-dimensional datasets and can model complex nonlinear decision boundaries through the kernel trick. Consequently, SVMs are widely applied in medical diagnosis tasks, including cancer subtype classification, ECG signal analysis, and medical image classification. The SVM decision function is defined as:

$$f(x) = \text{sign} \left(\sum_{i=1}^n \alpha_i y_i K(x_i, x) + b \right) \quad (2)$$

where:

- 1) $f(x)$ is the predicted class label for the input sample x .

- 2) $\text{sign}(\cdot)$ is the sign function that returns either +1 or -1 depending on which side of the decision boundary the sample lies.
- 3) x is the input feature vector to be classified.
- 4) x_i denotes the i^{th} support vector obtained during training.
- 5) n is the total number of support vectors.
- 6) α_i is the Lagrange multiplier associated with the i^{th} support vector, indicating its importance in defining the decision boundary.
- 7) $y_i \in \{-1, +1\}$ is the class label of the i^{th} support vector.
- 8) $K(x_i, x)$ is the kernel function that computes the similarity between the support vector x_i and the input sample x . Common kernel functions include the linear, polynomial, and radial basis function (RBF) kernels.
- 9) b is the bias (intercept) term that shifts the decision boundary.

Only the support vectors contribute to the final classification decision, making SVMs robust and effective for datasets with complex feature relationships.

Random Forests and ensemble methods such as **Gradient Boosting Machines, e.g., XGBoost and LightGBM** [8] have demonstrated strong performance in a wide range of healthcare prediction tasks. These algorithms effectively handle high-dimensional datasets, capture complex nonlinear relationships among variables, and provide feature-importance rankings that help identify the most influential clinical attributes. Random Forest constructs multiple decision trees using bootstrap samples and aggregates their predictions to improve accuracy and reduce overfitting. The Random Forest prediction for regression is computed as the average of the predictions from all decision trees:

$$\hat{y} = \frac{1}{B} \sum_{b=1}^B T_b(x) \quad (3)$$

where:

- 1) $\hat{y}$ is the final predicted output of the Random Forest model.
- 2) B is the total number of decision trees in the forest.
- 3) $T_b(x)$ is the prediction produced by the b^{th} decision tree for the input sample x .
- 4) b is the index of the decision tree, where $b = 1, 2, \dots, B$.
- 5) x is the input feature vector containing patient clinical attributes such as age, blood pressure, cholesterol level, ECG measurements, and other diagnostic features.

For classification tasks, the final prediction is typically determined by majority voting among all decision trees, whereas for regression tasks, the predictions are averaged. Gradient Boosting Machines (GBMs), including XGBoost and LightGBM, build predictive models sequentially by iteratively adding weak learners to correct the errors made by previous models. The prediction at the m^{th} iteration is given by:

$$F_m(x) = F_{m-1}(x) + \gamma_m h_m(x) \quad (4)$$

where:

- 1) $F_m(x)$ is the boosted model after the m^{th} iteration.
- 2) $F_{m-1}(x)$ is the prediction of the ensemble model from the previous iteration.
- 3) $h_m(x)$ is the weak learner (typically a shallow decision tree) trained at the m^{th} iteration to minimize the residual errors.
- 4) γ_m is the learning rate (or step size) that controls the contribution of the newly added weak learner.
- 5) m denotes the boosting iteration number.
- 6) x is the input feature vector.

Gradient Boosting algorithms optimize the prediction by sequentially minimizing a specified loss function, allowing them to achieve high predictive performance. Numerous studies have successfully applied Random Forest and Gradient Boosting methods for disease classification, risk prediction, and patient outcome forecasting, often achieving accuracies exceeding 85%. Their robustness to missing values, resistance to overfitting through ensemble learning, and ability to process mixed numerical and categorical data make them highly suitable for real-world clinical datasets.

k-Nearest Neighbors (k-NN) & Decision Trees: Both algorithms are widely used in healthcare due to their simplicity and interpretability, particularly when working with limited or moderately sized datasets. The k-Nearest Neighbors (k-NN) algorithm [9] is a non-parametric supervised learning method that classifies a new sample based on the labels of its nearest neighboring training samples. It is commonly applied in patient similarity analysis, disease subtype classification, and clinical decision support. Although k-NN is simple and intuitive, its computational cost increases with the size of the training dataset because all training samples must be examined during prediction. The classification rule is based on majority voting among the k nearest neighbors:

$$\hat{y} = \text{mode}\{y_i : i \in \mathcal{N}_k(x)\} \quad (5)$$

where:

- 1) $\hat{y}$ is the predicted class label of the input sample.
- 2) x is the input feature vector representing the patient data.
- 3) $\mathcal{N}_k(x)$ is the set of the k nearest training samples to x according to a selected distance metric (e.g., Euclidean distance).
- 4) k is the number of nearest neighbors considered during classification.
- 5) y_i is the class label of the i^{th} neighboring training sample.
- 6) $\text{mode}(\cdot)$ returns the most frequently occurring class label among the selected neighbors.

Decision Trees [10] are widely adopted in medical decision-making because they provide transparent and easily interpretable decision rules. They recursively partition the dataset by selecting features and threshold values that maximize information gain, thereby producing a tree structure that can

be readily interpreted by clinicians. The information gain used to evaluate a candidate split is calculated as:

$$\text{IG} = H(S) - \sum_{i \in \{L, R\}} \frac{|S_i|}{|S|} H(S_i) \quad (6)$$

where:

- 1) IG is the information gain obtained after splitting the dataset.
- 2) $H(S)$ is the entropy of the parent dataset S , which measures the impurity or uncertainty before splitting.
- 3) S is the parent dataset.
- 4) S_i represents the subset of samples after the split, where $i \in \{L, R\}$ corresponds to the left and right child nodes.
- 5) $|S|$ is the total number of samples in the parent dataset.
- 6) $|S_i|$ is the number of samples in the i^{th} child subset.
- 7) $H(S_i)$ is the entropy of the child subset after splitting.

A higher information gain indicates a better split because it results in a greater reduction in uncertainty. Decision Trees have been successfully applied to disease diagnosis, treatment recommendation, and risk assessment due to their ability to produce transparent decision paths that are easily understood by healthcare professionals.

B. Deep Learning Algorithms for Medical Applications

Deep learning algorithms has transformed medical data analysis by enabling automated feature extraction and hierarchical representation learning from complex datasets. Traditional machine learning methods that rely on custom features however DL models learn directly from raw medical data such as images, signals and electronic health records. The following architectures are among the most widely used in healthcare research and applications.

Convolutional Neural Networks (CNNs) & Recurrent Neural Networks (RNNs):

CNNs and RNNs are two fundamental deep learning architectures designed to process different types of medical data. Convolutional Neural Networks (CNNs) [11] are primarily used for extracting spatial features from medical images and have achieved remarkable success in disease detection, image segmentation, lesion localization, and medical image classification. They automatically learn hierarchical feature representations through convolutional operations, making them suitable for analyzing chest X-rays, magnetic resonance imaging (MRI), computed tomography (CT) scans, ultrasound images, and histopathological images. The convolution operation is defined as:

$$(F * K)(i, j) = \sum_m \sum_n F(i - m, j - n) K(m, n) \quad (7)$$

where:

- 1) $(F * K)(i, j)$ is the output feature map at spatial location (i, j) after convolution.
- 2) F is the input image or input feature map.

- 3) K is the convolution kernel (filter) used to extract specific features such as edges, textures, or patterns.
- 4) (i, j) denotes the spatial coordinates of the output feature map.
- 5) m and n are the indices of the kernel dimensions over which the convolution operation is performed.
- 6) $F(i - m, j - n)$ represents the input pixel or feature value corresponding to the kernel position.
- 7) $K(m, n)$ denotes the weight of the kernel at position (m, n) .

Recurrent Neural Networks (RNNs) [12] are designed to process sequential and temporal data by maintaining an internal memory through hidden states. This enables the network to capture dependencies between current and previous observations, making RNNs particularly suitable for analyzing time-series medical data such as electrocardiograms (ECG), electroencephalograms (EEG), electronic health records (EHRs), and patient monitoring signals. The hidden state of an RNN is computed as:

$$h_t = \sigma(W_{xh}x_t + W_{hh}h_{t-1} + b_h) \quad (8)$$

where:

- 1) h_t is the hidden state at the current time step t .
- 2) h_{t-1} is the hidden state from the previous time step.
- 3) x_t is the input vector at time step t .
- 4) W_{xh} is the weight matrix connecting the input vector to the hidden layer.
- 5) W_{hh} is the recurrent weight matrix connecting the previous hidden state to the current hidden state.
- 6) b_h is the bias vector of the hidden layer.
- 7) $\sigma(\cdot)$ is the activation function, typically the hyperbolic tangent (tanh) or sigmoid function.
- 8) t denotes the current time step in the input sequence.

CNNs effectively capture spatial hierarchies and local patterns in two-dimensional (2D) and three-dimensional (3D) medical images, making them highly effective for disease diagnosis, organ segmentation, and lesion detection. In contrast, RNNs model temporal dependencies in sequential medical data, enabling applications such as physiological signal analysis, disease progression prediction, patient monitoring, and clinical event forecasting.

Long Short-Term Memory (LSTM) Networks & Autoencoders: Long Short-Term Memory (LSTM) networks and Autoencoders are powerful deep learning architectures designed to learn complex patterns from medical data. LSTMs are an extension of Recurrent Neural Networks (RNNs) that overcome the vanishing gradient problem by introducing memory cells and gating mechanisms, enabling them to capture long-term dependencies in sequential data. Consequently, LSTMs are widely used for clinical time-series analysis, disease progression prediction, patient monitoring, and physiological signal processing such as electrocardiogram (ECG) and electroencephalogram (EEG) analysis. The LSTM cell is defined as:

$$\begin{aligned} f_t &= \sigma(W_f[h_{t-1}, x_t] + b_f), \\ C_t &= f_t \odot C_{t-1} + i_t \odot \tanh(W_C[h_{t-1}, x_t] + b_C), \\ h_t &= o_t \odot \tanh(C_t), \end{aligned} \quad (9)$$

where:

- 1) x_t is the input vector at time step t .
- 2) h_t is the hidden state at the current time step.
- 3) h_{t-1} is the hidden state from the previous time step.
- 4) C_t is the cell state (long-term memory) at the current time step.
- 5) C_{t-1} is the cell state from the previous time step.
- 6) f_t is the forget gate, which determines how much information from the previous cell state should be retained.
- 7) i_t is the input gate, which controls how much new information is written into the cell state.
- 8) o_t is the output gate, which determines how much information from the cell state is passed to the hidden state.
- 9) W_f is the weight matrix associated with the forget gate.
- 10) W_C is the weight matrix used to compute the candidate cell state.
- 11) b_f is the bias vector of the forget gate.
- 12) b_C is the bias vector of the candidate cell state.
- 13) $[h_{t-1}, x_t]$ denotes the concatenation of the previous hidden state and the current input vector.
- 14) $\sigma(\cdot)$ is the sigmoid activation function, which produces values between 0 and 1.
- 15) $\tanh(\cdot)$ is the hyperbolic tangent activation function, producing values between -1 and 1 .
- 16) $\odot$ denotes element-wise multiplication.
- 17) t represents the current time step in the input sequence.

Autoencoders are unsupervised neural networks that learn compact and meaningful feature representations by encoding the input into a lower-dimensional latent space and reconstructing the original input from this compressed representation. They are widely used for feature extraction, dimensionality reduction, anomaly detection, and medical image denoising. The encoding and decoding operations are expressed as:

$$z = \sigma(W_e x + b_e), \quad \hat{x} = \sigma(W_d z + b_d) \quad (10)$$

where:

- 1) x is the original input feature vector or image.
- 2) z is the latent (compressed) representation learned by the encoder.
- 3) $\hat{x}$ is the reconstructed output produced by the decoder.
- 4) W_e is the encoder weight matrix.
- 5) b_e is the encoder bias vector.
- 6) W_d is the decoder weight matrix.
- 7) b_d is the decoder bias vector.
- 8) $\sigma(\cdot)$ is the activation function used in the encoder and decoder layers.

LSTMs effectively model long-term temporal dependencies in sequential medical data, making them suitable for

disease progression prediction, patient monitoring, and physiological signal analysis. Autoencoders learn compact feature representations that enable efficient dimensionality reduction, medical image denoising, anomaly detection, and unsupervised feature learning in healthcare applications [13, 14].

U-Net & Residual Networks (ResNet): U-Net and ResNet are two widely adopted convolutional neural network architectures with distinct strengths in medical image analysis. U-Net is specifically designed for biomedical image segmentation and employs a symmetric encoder–decoder architecture with skip connections that preserve fine-grained spatial information. These skip connections combine high-resolution feature maps from the encoder with the upsampled feature maps of the decoder, enabling accurate localization of anatomical structures and lesions. The decoder operation at the l^{th} layer is expressed as:

$$x_{\text{decoder}}^{(l)} = \text{Conv} \left(\text{Concat} \left(x_{\text{encoder}}^{(L-l)}, \text{Up} \left(x_{\text{decoder}}^{(l-1)} \right) \right) \right) \quad (11)$$

where:

- 1) $x_{\text{decoder}}^{(l)}$ is the output feature map of the decoder at layer l .
- 2) $x_{\text{encoder}}^{(L-l)}$ is the corresponding feature map from the encoder connected through the skip connection.
- 3) $x_{\text{decoder}}^{(l-1)}$ is the output feature map from the previous decoder layer.
- 4) L is the total number of encoder layers.
- 5) l denotes the current decoder layer.
- 6) $\text{Up}(\cdot)$ is the upsampling operation that increases the spatial resolution of the decoder feature map.
- 7) $\text{Concat}(\cdot)$ represents the channel-wise concatenation of encoder and decoder feature maps.
- 8) $\text{Conv}(\cdot)$ denotes the convolution operation that extracts refined features after concatenation.

ResNet introduces residual (identity) connections that allow information to bypass one or more convolutional layers, thereby mitigating the vanishing gradient problem and enabling the successful training of very deep neural networks. The residual learning function is expressed as:

$$y = \mathcal{F}(x, \{W_i\}) + x \quad (12)$$

where:

- 1) y is the output feature map of the residual block.
- 2) x is the input feature map, which also serves as the identity shortcut connection.
- 3) $\mathcal{F}(x, \{W_i\})$ is the residual mapping learned by the stacked convolutional layers.
- 4) $\{W_i\}$ denotes the set of learnable weight matrices associated with the convolutional layers in the residual block.
- 5) i represents the index of the convolutional layer within the residual block.

The encoder–decoder architecture of U-Net enables precise localization of organs, tissues, and tumors, making it highly effective for medical image segmentation. In contrast, the residual connections in ResNet facilitate feature reuse and efficient gradient propagation, thereby improving convergence, classification accuracy and lesion detection performance in deep neural networks.[15]

Vision Transformers (ViTs): Vision Transformers (ViTs) are transformer-based deep learning architectures that utilize self-attention mechanisms to model long-range dependencies within images. Instead of processing the entire image directly, ViTs divide an image into a sequence of fixed-size patches, embed these patches into feature vectors, and learn relationships among them using the transformer encoder. The scaled dot-product self-attention mechanism is defined as:

$$\text{Attention}(Q, K, V) = \text{softmax} \left(\frac{QK^T}{\sqrt{d_k}} \right) V \quad (13)$$

where:

- 1) Q is the query matrix derived from the embedded image patches.
- 2) K is the key matrix representing the encoded features of the image patches.
- 3) V is the value matrix containing the feature representations of the image patches.
- 4) K^T denotes the transpose of the key matrix.
- 5) d_k is the dimensionality of the key vectors used to normalize the attention scores.
- 6) $\frac{QK^T}{\sqrt{d_k}}$ computes the scaled similarity scores between the query and key vectors.
- 7) $\text{softmax}(\cdot)$ converts the similarity scores into normalized attention weights.
- 8) $\text{Attention}(Q, K, V)$ represents the weighted combination of the value vectors based on the computed attention weights.

By learning relationships among all image patches simultaneously, Vision Transformers effectively capture global contextual information that may not be fully captured by conventional convolutional neural networks. Consequently, ViTs have demonstrated excellent performance in medical image classification, disease detection, organ segmentation, and computer-aided diagnosis, particularly when trained on large-scale datasets or integrated with transfer learning techniques. [16]

C. Hybrid and Ensemble Approaches

Hybrid and ensemble approaches combine multiple algorithms to enhance accuracy and generalization in healthcare applications and they are especially effective for heterogeneous medical datasets where single model may miss key patterns, leveraging complementary features and decision strategies from traditional and deep learning methods.

Graph Neural Networks with Transformers combine the relational reasoning capabilities of Graph Neural Networks

(GNNs) [17] with the powerful attention mechanisms of transformers. The core attention mechanism is defined as:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \quad (14)$$

where:

- 1) Q is the query matrix derived from the input node features.
- 2) K is the key matrix representing the encoded features of the graph nodes.
- 3) V is the value matrix containing the feature representations of the graph nodes.
- 4) K^T denotes the transpose of the key matrix.
- 5) d_k is the dimensionality of the key vectors used to normalize the attention scores.
- 6) $\frac{QK^T}{\sqrt{d_k}}$ computes the scaled similarity scores between the query and key vectors.
- 7) $\text{softmax}(\cdot)$ converts the similarity scores into normalized attention weights.
- 8) $\text{Attention}(Q, K, V)$ represents the weighted combination of the value vectors based on the computed attention weights.

Graph Neural Networks with Transformers integrate local neighborhood information through graph connectivity with global contextual information through self-attention. Consequently, these hybrid models effectively capture complex relational dependencies in biomedical networks, enabling accurate analysis of drug–drug interactions, protein–protein interaction networks, healthcare knowledge graphs, and other medical graph-structured data.

Autoencoder Random Forest Ensembles combine the feature learning capabilities of autoencoders with the interpretability and robustness of Random Forest classifiers. Initially, the autoencoder learns a compressed representation of the input data through the encoder:

$$z = f_e(x) = \sigma(W_e x + b_e) \quad (15)$$

where:

- 1) x is the input feature vector.
- 2) z is the latent (compressed) representation learned by the encoder.
- 3) $f_e(\cdot)$ is the encoder function that maps the input into the latent space.
- 4) W_e is the encoder weight matrix.
- 5) b_e is the encoder bias vector.
- 6) $\sigma(\cdot)$ is the activation function used by the encoder.

The compressed representation is then used as the input to the Random Forest classifier:

$$\hat{y} = \frac{1}{B} \sum_{b=1}^B T_b(z) \quad (16)$$

where:

- 1) $\hat{y}$ is the final predicted output of the Random Forest ensemble.
- 2) B is the total number of decision trees in the ensemble.
- 3) $T_b(z)$ is the prediction of the b^{th} decision tree for the latent representation z .
- 4) z is the latent representation provided by the autoencoder.
- 5) b denotes the index of the decision tree in the ensemble.

By combining automatic feature extraction with ensemble learning, Autoencoder Random Forest ensembles effectively reduce feature dimensionality while maintaining robust and interpretable classification performance. Consequently, they have demonstrated strong performance in medical image classification, electronic health record analysis, disease prediction, and other healthcare applications. [18]

Multi-modal Fusion Networks integrate complementary information from multiple healthcare data sources, such as medical images, clinical notes, and laboratory test results. Separate encoders are employed to extract feature representations from each data modality:

$$\begin{aligned} z_{\text{img}} &= f_{\text{img}}(x_{\text{image}}) \\ z_{\text{text}} &= f_{\text{text}}(x_{\text{clinical}}) \\ z_{\text{lab}} &= f_{\text{lab}}(x_{\text{lab}}) \end{aligned} \quad (17)$$

where:

- 1) x_{image} is the input medical image.
- 2) x_{clinical} is the input clinical text or electronic health record.
- 3) x_{lab} is the input laboratory test data.
- 4) $f_{\text{img}}(\cdot)$ is the encoder that extracts features from medical images.
- 5) $f_{\text{text}}(\cdot)$ is the encoder that extracts features from clinical text.
- 6) $f_{\text{lab}}(\cdot)$ is the encoder that extracts features from laboratory data.
- 7) z_{img} is the latent feature representation of the medical image.
- 8) z_{text} is the latent feature representation of the clinical text.
- 9) z_{lab} is the latent feature representation of the laboratory data.

The extracted representations are subsequently combined using different fusion strategies:

$$z_{\text{fused}} = \begin{cases} f_{\text{early}}(x_{\text{image}}, x_{\text{clinical}}, x_{\text{lab}}), & \text{(early fusion)} \\ f_{\text{intermediate}}(z_{\text{img}}, z_{\text{text}}, z_{\text{lab}}), & \text{(intermediate fusion)} \\ f_{\text{late}}(y_{\text{img}}, y_{\text{text}}, y_{\text{lab}}), & \text{(late fusion)} \end{cases} \quad (18)$$

where:

- 1) z_{fused} is the final fused feature representation.
- 2) $f_{\text{early}}(\cdot)$ is the early fusion function that combines the raw input modalities.

- 3) $f_{\text{intermediate}}(\cdot)$ is the intermediate fusion function that combines the latent feature representations.
- 4) $f_{\text{late}}(\cdot)$ is the late fusion function that combines the prediction outputs from each modality.
- 5) y_{img} , y_{text} , and y_{lab} are the prediction outputs of the image, clinical text, and laboratory encoders, respectively.

By integrating heterogeneous healthcare data sources, Multi-modal Fusion Networks provide a comprehensive representation of patient information. Consequently, these architectures have demonstrated superior performance in disease diagnosis, patient risk stratification, clinical decision support, personalized treatment recommendation, and comprehensive patient profiling. [19]

Ensemble of Deep Learning Models combine the predictions of multiple deep learning architectures, such as Convolutional Neural Networks (CNNs), Vision Transformers (ViTs), and Residual Networks (ResNets), to improve prediction accuracy and robustness. The ensemble prediction using simple averaging is expressed as:

$$\hat{y} = \frac{1}{N} \sum_{i=1}^N f_i(x) \quad (19)$$

where:

- 1) $\hat{y}$ is the final ensemble prediction.
- 2) N is the total number of deep learning models in the ensemble.
- 3) $f_i(x)$ is the prediction of the i^{th} deep learning model for the input x .
- 4) x is the input sample, such as a medical image or patient data.
- 5) i denotes the index of the deep learning model in the ensemble.

Alternatively, weighted averaging assigns different importance to individual models:

$$\hat{y} = \sum_{i=1}^N w_i f_i(x), \quad \sum_{i=1}^N w_i = 1 \quad (20)$$

where:

- 1) $\hat{y}$ is the weighted ensemble prediction.
- 2) w_i is the weight assigned to the prediction of the i^{th} model.
- 3) $f_i(x)$ is the prediction of the i^{th} deep learning model.
- 4) N is the total number of models in the ensemble.
- 5) $\sum_{i=1}^N w_i = 1$ ensures that the weights are normalized and their sum equals one.

By combining the strengths of multiple deep learning architectures, ensemble methods reduce model bias and variance while improving prediction robustness and generalization. Consequently, these approaches have demonstrated superior performance in medical image classification, disease diagnosis, lesion detection, and rare disease identification compared with individual deep learning models. Overall, deep learning and hybrid architectures enable intelligent healthcare systems

for early disease detection, accurate diagnosis, and personalized treatment planning. Table I compares the key machine learning, deep learning, and hybrid algorithms discussed in this section. [20]

D. Performance Metrics and Evaluation Frameworks

The evaluation of AI models in healthcare requires comprehensive frameworks assessing statistical performance, clinical reliability, and real-world utility. This section outlines essential metrics and validation methodologies for benchmarking AI systems across medical applications, including classification, segmentation, regression, and survival analysis tasks as detailed in Table II. These metrics span multiple dimensions from fundamental statistical measures to advanced clinical impact assessments, ensuring robust evaluation across diverse healthcare scenarios.

1) Standard Classification Metrics: Performance in classification is not limited to accuracy; extends to precision-recall trade-offs for clinical contexts. The specificity is important to reduce the false positive rate in screening applications, whereas high sensitivity is important to reduce the false negative rate in diagnostic confirmation applications, as illustrated in Table II. The F1-score balances these competing needs, and AUC metrics provide comprehensive threshold-agnostic performance assessment across diverse clinical scenarios. Additional metrics like Matthews Correlation Coefficient offer robust evaluation for imbalanced datasets commonly encountered in rare disease detection and medical diagnostics.

2) Medical Imaging and Segmentation Metrics: Medical image analysis employs spatial accuracy metrics quantifying anatomical alignment and boundary precision. Dice and IoU scores evaluate volumetric overlap for treatment planning, while Hausdorff Distance ensures surgical safety margins as summarized in Table II. Additional metrics like Average Surface Distance and Volumetric Similarity provide comprehensive spatial evaluation for precise anatomical modeling. These metrics are particularly crucial for radiotherapy planning, surgical navigation, and quantitative disease progression monitoring in longitudinal studies.

3) Diagnostic Performance Assessment: Clinical validation requires metrics translating algorithmic performance into diagnostic confidence. Predictive values (PPV, NPV) offer patient-specific probability estimates, while likelihood ratios quantify diagnostic impact as detailed in Table II. Calibration metrics ensure probability outputs match observed event rates, crucial for reliable risk prediction models in clinical practice.

4) Statistical Validation and Clinical Utility: Robust validation ensures model reliability through cross-validation, bootstrapping, and external validation techniques outlined in Table II. Clinical utility metrics assess tangible benefits in healthcare workflows, including time efficiency, inter-rater

TABLE I: Comparison of Traditional and Deep Learning Algorithms in Healthcare

Algorithm	Key Strengths	Common Healthcare Applications
Machine Learning Algorithms		
LR	Interpretable, strong statistical basis	Clinical risk prediction, diagnostic models
RF	Handles high dimensional data, robust, feature importance	Disease classification, outcome forecasting
GB	High accuracy, models complex patterns	Patient outcome prediction
SVM	Effective for high dimensional data, complex boundaries	Medical image analysis, cancer subtype classification
k-NN	Simple, intuitive, good for pattern recognition	Patient similarity analysis, disease subtype classification
DT	Transparent, easily interpretable, simple validation	Clinical guidelines, patient stratification
Deep Learning Algorithm		
CNN	Captures spatial hierarchies, strong in image feature extraction	Medical image classification, disease detection
RNN	Models sequential dependencies in temporal data	ECG/EEG analysis, patient monitoring
LSTM	Learns long-term dependencies, handles vanishing gradients	Outcome prediction, clinical time series forecasting
U-Net	Encoder-decoder with skip connections, precise localization	Tumor boundary detection
ResNet	Skip connections enable deeper networks and stability	Disease classification, lesion detection
Autoencoder	Unsupervised feature learning, data reconstruction	Anomaly detection, image denoising
ViT	Uses self attention for global feature learning	Histopathology, radiology, medical image classification
Hybrid and Esemble Algorithm		
CNN-RNN	Combines spatial and temporal analysis capabilities	Cardiac MRI analysis, video based surgical analytics
GNN-Transformer	Integrates relational reasoning with attention mechanisms	Drug interaction prediction, biomedical network analysis
Autoencoder-RF	Feature learning with interpretable decision making	Genomic data analysis, EHR processing
Multi modal Fusion	Integrates diverse data sources comprehensively	Personalized treatment, multi parameter diagnostics
DL Ensemble	Mitigates individual model biases, improves robustness	Medical imaging diagnostics, rare disease detection
Neuro-Symbolic AI	Combines pattern recognition with logical reasoning	Clinical decision support

Note: LR = Logistic Regression; RF = Random Forests; GB = Gradient Boosting; SVM = Support Vector Machines; k-NN = k-Nearest Neighbors; DT = Decision Trees; CNN = Convolutional Neural Network; RNN = Recurrent Neural Network; LSTM = Long Short-Term Memory; U-Net = U-shaped Network; ResNet = Residual Network; ViT = Vision Transformer; GNN = Graph Neural Network; EHR = Electronic Health Record; DL = Deep Learning.

reliability, and net benefit analysis for informed clinical decision-making and resource allocation.

IV. AI APPLICATIONS IN HEALTHCARE DOMAINS

A. Cardiovascular Diseases

A heart disease prediction system using decision tree classifiers evaluated ID3, C4.5, C5.0, and J48 models, with J48 achieving the highest accuracy of 99.62%, while the Maximal Frequent Itemset Algorithm combined with C4.5 achieved 92% accuracy, 0.82 precision, and 0.92 recall [21]. A comparative analysis of machine learning models for heart disease prediction found that the Support Vector Machine (SVM) achieved an accuracy of 95.34%, with age, cholesterol, and maximum heart rate identified as the most influential predictive features [22]. An explainable artificial intelligence (XAI)-based heart disease prediction system utilizing LightGBM and XGBoost models achieved an accuracy of 91.94% using clinical and lifestyle data from more than 300,000 patients [23].

A coronary artery disease prediction pipeline incorporating Bald Eagle Search Optimization (BESO) for feature selection achieved an accuracy of 92% using a Random Forest classifier [24]. An explainable cardiovascular disease prediction system based on a Random Forest classifier achieved an accuracy of 98% by integrating global SHAP and local LIME to provide both overall model interpretability and patient-specific explanations [25]. A comparative study found that the KNN classifier achieved the highest accuracy of 92%, outperforming Random Forest and SVM models [26]. Another intelligent healthcare system reported a Random Forest accuracy of 99.89% for heart attack prediction [27].

A heart failure prediction model comparing CatBoost, Logistic Regression, and KNN classifiers was evaluated using the F1-score and ROC curve to assess predictive performance [28]. A machine learning-based heart disease prediction model evaluated Decision Tree, KNN, Naive Bayes, Random Forest, and XGBoost classifiers, with XGBoost achieving the highest accuracy of 93% [29]. A comparative evaluation of 12 machine learning models identified Gradient Boosting Machine as the best-performing classifier with an accuracy of 90.20%,

TABLE II: Comprehensive Framework for Healthcare AI Performance Evaluation

Category	Metric	Formula	Clinical Interpretation	Primary Use Case
Classification	Accuracy	$\frac{TP+TN}{TP+TN+FP+FN}$	Overall diagnostic correctness	General screening
	Precision	$\frac{TP}{TP+FP}$	Reliability of positive diagnosis (>90%)	Confirmatory testing
	Recall	$\frac{TP}{TP+FN}$	Ability to detect true cases (>95%)	Disease screening
	F1-Score	$2 \times \frac{P \times R}{P+R}$	Balance of precision/recall trade-off	Imbalanced data
	AUC-ROC	Area under curve	Overall discrimination ability (>0.90)	Model comparison
Segmentation	Dice Score	$\frac{2 P \cap G }{ P + G }$	Volumetric overlap accuracy (>0.90)	Tumor segmentation
	IoU	$\frac{ P \cap G }{ P \cup G }$	Spatial overlap measure (>0.85)	Organ delineation
	Hausdorff Dist.	$\max(d(\partial P, \partial G))$	Worst-case boundary error (<5mm)	Surgical planning
Regression	MAE	$\frac{1}{n} \sum y_i - \hat{y}_i $	Average prediction error	Continuous outcomes
	RMSE	$\sqrt{\frac{1}{n} \sum (y_i - \hat{y}_i)^2}$	Penalized large errors	Risk scores
	R ²	$1 - \frac{\sum (y_i - \hat{y}_i)^2}{\sum (y_i - \bar{y})^2}$	Variance explained (>0.80)	Predictive models
	MAPE	$\frac{100\%}{n} \sum \left \frac{y_i - \hat{y}_i}{y_i} \right $	Percentage error	Clinical measurements
Diagnostic	PPV/NPV	$\frac{TP}{TP+FP}, \frac{TN}{TN+FN}$	Post-test probability (prevalence-dependent)	Clinical decision making
	Likelihood Ratio	$\frac{Sen}{1-Spec}$	Diagnostic impact strength (>10 strong)	Evidence-based medicine
	Cohen's κ	Agreement correction	Expert concordance (>0.80)	Validation studies
Validation	K-Fold CV	$\frac{1}{K} \sum_{i=1}^K Perf_i$	Generalizability estimate (stable folds)	Model development
	Bootstrapping CI	$\hat{\theta} \pm z \cdot SE$	Performance uncertainty (narrow CI)	Clinical adoption
	Stratified Sampling	Class balance preservation	Population representation	Medical datasets
Clinical Impact	Net Benefit	$TPR - \frac{FP \times \theta}{1-\theta}$	Risk-benefit trade-off (positive value)	Decision curve analysis
	Time Efficiency	$T_{std} - T_{AI}$	Workflow improvement (>30% reduction)	Clinical integration
	Inter-rater Rel.	Fleiss' κ /ICC	Multi-expert consensus (>0.75)	Multi-center trials
Survival Ana	C-index	Concordance probability	Predictive accuracy for time-to-event	Cancer prognosis
	Brier Score	$\frac{1}{N} \sum (p_i - o_i)^2$	Probability calibration (<0.25 good)	Risk prediction models
	Integrated Brier	$\frac{1}{\max(t)} \int_0^{\max(t)} BS(t)dt$	Time-dependent calibration	Survival models
	Calibration Slope	Slope of calibration plot	Calibration quality (1.0 ideal)	Risk models
Fairness & Bias	Equalized Odds	$P(\hat{Y} = 1 Y = y, A = a) = P(\hat{Y} = 1 Y = y, A = b)$	Fairness across subgroups	Equity assessment
	Demographic Parity	$P(\hat{Y} = 1 A = a) = P(\hat{Y} = 1 A = b)$	Equal positive rate	Bias detection
	Disparate Impact	$\frac{P(\hat{Y}=1 A=minority)}{P(\hat{Y}=1 A=majority)}$	Ratio of positive rates (>0.8)	Regulatory compliance
	AUC Difference	$ AUC_a - AUC_b $	Performance gap (<0.05)	Subgroup analysis

Note: Category abbreviations: Class. = Classification, Seg. = Segmentation, Diag. = Diagnostic Assessment, Val. = Statistical Validation, Clin. = Clinical Impact. Metric abbreviations: TP/FP/TN/FN = True/False Positive/Negative, P/R = Precision/Recall, AUC-ROC = Area Under ROC Curve, IoU = Intersection over Union, PPV/NPV = Positive/Negative Predictive Value, CI = Confidence Interval, CV = Cross-Validation, TPR = True Positive Rate, θ = risk threshold, ICC = Intraclass Correlation Coefficient, Sen = Sensitivity, Spec = Specificity.

followed by CatBoost (89.71%) and LightGBM (88.73%) [30].

Another study using a preprocessed Cleveland heart disease dataset identified 11 important clinical features and evaluated Support Vector Machine, Gaussian Naïve Bayes, Decision Tree, Artificial Neural Network, and Logistic Regression classifiers for heart disease prediction [31]. A comparison between WEKA and Google Colab (Scikit-Learn) showed that WEKA consistently achieved higher classification accuracies, with SVM and Logistic Regression reaching 90.67% compared to 76.58% in Scikit-Learn [32]. A Convolutional Neural Network (CNN)-based framework for cardiovascular disease prediction achieved 98.54% accuracy, 98.56% F1-score, 98.60% AUC-ROC, and 99.00% specificity, outperforming Logistic Regression, SVM, and Gradient Boosting classifiers on the Heart Disease dataset [33].

B. Neurological Disorders and Brain Analysis

A comparative evaluation of machine learning methods for neurological disorder prediction found that Gradient Boosting achieved 89% accuracy for early Parkinson's disease detection, while K-Nearest Neighbors (KNN) attained 85% accuracy for epilepsy detection using EEG data [34]. A Random Forest-based model for Parkinson's disease prediction achieved 93% accuracy using clinical and demographic data, while SHAP-based explainable AI techniques provided interpretable insights into the model's predictions [35]. A logistic regression-based clinical prediction model for neonatal purulent meningitis achieved an AUC of 0.765, providing an effective tool for early risk assessment in septic neonates [36].

A LASSO-based model using six key clinical features, including cerebrospinal fluid (CSF) pressure and India ink staining, achieved AUCs of 0.919 and 0.921 in the training and validation cohorts, respectively, for distinguishing cryptococcal meningitis from tuberculous meningitis [37]. A machine learning model for predicting adverse outcomes in full-term neonates with bacterial meningitis identified Logistic Regression as the best-performing classifier, achieving 89.0% accuracy and an AUC of 0.908 [38]. Another logistic regression-based nomogram identified low birth weight, elevated procalcitonin, cesarean delivery, and premature membrane rupture as significant risk factors for purulent meningitis in very preterm infants. Additionally, CNN architectures, including U-Net and ResNet, achieved up to 93.8% accuracy for predicting genetic biomarkers from multi-parametric MRI, although MGMT promoter methylation prediction remained limited with an AUROC of 0.62 [39].

A comparative evaluation of deep learning models for multiclass brain tumor classification using MRI found that EfficientNet achieved the highest accuracy of 98.72%, followed by the Swin Transformer with 98.08%, outperforming conventional CNN models [40]. A comparative evaluation of CNN architectures on a public MRI dataset found that ResNet50 achieved the highest accuracy of 95.35%, followed by VGG16 with 93.93%, while U-Net attained 79% accuracy, highlighting the importance of model architecture in medical image classification [41].

C. Oncology and Cancer Diagnosis

Machine learning algorithms have been widely applied for breast cancer diagnosis and prognosis. A comparative evaluation of eight machine learning classifiers identified Logistic Regression as the best-performing model with 91.67% accuracy, while feature selection improved the performance of LightGBM to 90.74% [42]. A comparative analysis on the Wisconsin Diagnostic Breast Cancer dataset found that Random Forest achieved the highest accuracy of 92.64% after feature selection and hyperparameter tuning [43]. A breast cancer survival prediction model using Random Forest, SVM, and Multilayer Perceptron achieved the best performance with Random Forest, obtaining 96% accuracy and 93% AUC [44]. A multimodal artificial intelligence system integrating clinical, mammography, and ultrasound data achieved 90.1% accuracy for breast cancer risk stratification, demonstrating performance comparable to radiologists [45].

A comparative study of seven classifiers on the Wisconsin Breast Cancer dataset identified SVM as the best-performing model with an accuracy of 97.66% [46]. Feature selection was shown to improve the performance of LightGBM for breast cancer classification [47]. Support Vector Machine was reported as the most effective classifier for distinguishing benign and malignant breast tumors [48]. Decision tree-based models achieved up to 93.56% accuracy after appropriate data preprocessing and feature selection [49, 50].

A Logistic Regression-based breast cancer prediction model achieved 97% accuracy while maintaining high computational efficiency [51]. A Random Forest model using transcriptomic data achieved 93.6% accuracy and 91.3% AUC for predicting distant breast cancer metastasis [52]. A comparative study found that K-Nearest Neighbors (KNN) achieved the highest accuracy of 98.4%, outperforming Random Forest and other classifiers for breast cancer prediction [53]. Random Forest and SVM were identified as the most effective classifiers when combined with feature selection and cross-validation, while KNN exhibited the shortest training time [54].

D. Dermatology and Skin Diseases

Machine learning and deep learning techniques have been widely applied for skin disease diagnosis. A Hierarchical Prototypical Decision Tree (HPDT) integrating deep learning embeddings, prototypical networks, and decision trees achieved a hierarchical F1-score of 81.56 and an accuracy of 60.87% on 235,268 dermoscopic images [55]. An ensemble model based on transfer learning using ResNet50 and Logistic Regression achieved 96% accuracy for classifying five pediatric skin diseases [?].

A comparative evaluation of six classifiers identified Logistic Regression and Stochastic Gradient Descent (SGD) as the best-performing models, achieving accuracies exceeding 98% after feature selection [56]. A comprehensive review reported that deep learning models, including CNNs, transformers, and ensemble methods, achieved dermatologist-level performance with accuracies exceeding 98% [57]. A hybrid ensemble machine learning (HEML) framework combining

image preprocessing with Decision Tree, KNN, and SVM classifiers achieved an accuracy of 95.50%, outperforming individual classifiers [58].

A comprehensive survey reported that image-based skin disease diagnosis achieved accuracies ranging from 50% to 100%, while tissue-based feature methods consistently reached approximately 94% accuracy [59]. A comparative study of KNN, SVM, and Decision Tree classifiers found that SVM achieved 95% accuracy on the ISIC 2019 dataset and 97% on the HAM10000 dataset [60]. A comparative evaluation identified SVM as the best-performing classifier with an accuracy of 98.8%, while KNN achieved the highest sensitivity (91%) and specificity (99%) [61].

E. Ophthalmology and Vision Disorders

Deep learning-based cataract diagnosis using fundus images achieved an accuracy of 97.62% through a Dual-Stream Cataract Evaluation Network (DCEN) based on ResNet architectures [62]. A CNN-based CataractNet model outperformed several pre-trained CNN architectures for cataract detection from fundus images while requiring lower computational time [63]. A YOLOv8-based framework for ophthalmic disease classification using fundus images demonstrated effective disease detection by achieving high accuracy, precision, recall, and F1-score across 5,887 retinal images [64].

A smartphone-based cataract detection system using a CNN achieved an accuracy of 98.79% and enabled automated cataract screening from mobile eye images [65]. A mobile application based on a 16-layer CNN effectively detected and classified cataracts from fundus images [66]. A back-propagation neural network using luminance and texture features from fundus images demonstrated effective cataract classification [67]. A smartphone-based cataract detection application employing a Support Vector Machine (SVM) achieved 95% accuracy, demonstrating the feasibility of accessible AI-assisted diagnosis [68].

A smartphone application integrating image processing with KNN, SVM, and Naïve Bayes classifiers identified KNN as the best-performing model with an accuracy of 83.07% [69]. An SVM-based framework for nuclear cataract detection achieved 95.2% accuracy using retinal blood vessel features, with the Radial Basis Function (RBF) kernel providing the best performance [70]. A mobile cataract detection system using Gray Level Co-occurrence Matrix (GLCM) features and K-Nearest Neighbors (KNN) achieved a precision of 97.5%, with $k = 1$ providing the best classification performance [71].

F. Hepatology and Liver Diseases

Machine learning and deep learning techniques have been extensively applied for liver disease diagnosis and prediction. A comparative evaluation on the BUPA and Indian Liver Patient Dataset (ILPD) found that a CNN outperformed conventional machine learning models, achieving accuracies of 75.55% and 72.00%, respectively [42].

A deep learning framework based on a Stacked Sparse Autoencoder (SSAE) achieved 97.2% accuracy for liver disease

classification from ultrasound images [72]. A deep learning model for liver tumor detection and classification in CT images achieved a Dice similarity coefficient of 98.59%, demonstrating excellent segmentation performance [73]. A two-stage deep learning framework for nonalcoholic fatty liver disease (NAFLD) detection achieved AUROC values of 0.90 for NAFLD and 0.93 for severe NAFLD using ultrasound images [74].

A comparative evaluation of machine learning algorithms identified LightGBM and Random Forest as the best-performing models, both achieving an accuracy of 63% on the Indian Liver Patient Dataset [75]. A comparative study on 32,000 patient records found that Random Forest achieved the highest accuracy of 97.3% for liver disease prediction [76]. A comparison of ten machine learning algorithms identified Logistic Regression as the best-performing model, achieving 72.8% accuracy and an AUC of 0.806 [77]. An ensemble model combining SVM, Random Forest, Decision Tree, and XGBoost achieved 95.24% accuracy with a ROC value of 99.08% [78].

A machine learning model for predicting drug-induced hepatic steatosis achieved the highest ROC-AUC using SVM, reaching 0.975 for the rat *in vitro* model and 0.966 for the rat *in vivo* model [79]. A hybrid ensemble model incorporating SHAP and LIME achieved 98.38% accuracy for liver disease prediction while providing interpretable clinical insights [80]. A Tree Selection and Stacking Ensemble based on Random Forest achieved 99.92% accuracy for liver disease diagnosis, with SHAP and LIME used to identify clinically relevant biomarkers [81].

G. Nephrology and Kidney Diseases

Machine learning and deep learning techniques have been widely applied for kidney disease and kidney stone detection. A hybrid CNN-ResNet framework for kidney stone detection combined CNN-based feature extraction with ResNet to improve diagnostic accuracy using ultrasound and CT images [82]. A comparative evaluation of pre-trained CNN architectures identified Inception-V3 as the best-performing model, achieving 98.52% accuracy for kidney stone detection from abdominal CT images [83].

A CNN-based point-of-care system using smartphone microscopy achieved 88% accuracy and an average F1-score of 0.84 for classifying four types of kidney stones [84]. An ensemble deep learning model based on fine-tuned VGG-16 achieved 96% accuracy, precision, recall, and F1-score for classifying kidney CT scans into cyst, normal, stone, and tumor categories [85]. A transfer learning framework using InceptionV3, VGG16, ResNet50, and CNN AlexNet achieved 99.96% accuracy and a perfect ROC-AUC of 1.0 for classifying kidney CT images into cyst, normal, stone, and tumor categories [86].

A comparative evaluation of six machine learning algorithms identified Logistic Regression as the best-performing model for kidney stone prediction, achieving an accuracy of 78% [87]. An automated kidney stone detection system based

on ultrasound image processing achieved 96.82% accuracy and 92.16% sensitivity using median filtering, adaptive thresholding, and morphological analysis [88]. A comparative study of ten machine learning algorithms for kidney stone prediction using urine analysis data found that Logistic Regression achieved the highest accuracy of 93% with 87% precision [89].

H. Endocrinology and Metabolic Disorders

Machine learning techniques have been extensively applied for early diabetes prediction. A comparative evaluation of KNN and Naïve Bayes on the Pima Indians Diabetes Dataset employed data preprocessing and 10-fold cross-validation, with model performance assessed using confusion matrix metrics [90]. A Random Forest-based model for early diabetes prediction achieved an accuracy of 87% [91].

A modified Genetic J48 decision tree achieved 95.8% accuracy and 97.2% efficiency for diabetes prediction by eliminating irrelevant features [92]. A comparative study evaluated KNN, Logistic Regression, Decision Tree, SVM, Gradient Boosting, and Random Forest classifiers for early diabetes prediction using the Pima Indians Diabetes Dataset [93]. A machine learning framework combining data preprocessing, association rule mining, Artificial Neural Networks (ANN), Random Forest, and K-means clustering demonstrated effective early diabetes prediction [94].

A machine learning system utilizing Decision Tree, ANN, Naïve Bayes, and SVM classifiers enabled early diabetes prediction from patient clinical data [95]. A Decision Tree J48 model effectively predicted diabetes in pregnant women using eight clinical features from the Pima Indians Diabetes Dataset with low computational cost [96]. A decision tree model optimized using Gini impurity identified glucose, BMI, and age as the most significant predictors for diabetes while providing interpretable decision rules [97].

I. Other Medical Specialties

A comprehensive review highlighted the effectiveness of deep learning models, including CNNs and Transformers, for medical image analysis, personalized treatment, and clinical decision support across diverse healthcare applications [98]. A systematic review of 441 studies on MRI-based neurological disorder diagnosis reported that CNNs, ResNet, VGG, and SVM were the most widely used models, with some studies achieving accuracies of up to 99% [99]. A computer-aided diagnosis system combining a customized VGG-16 model with SVM achieved 97.51% accuracy, 97.77% specificity, and 96.5% sensitivity for pneumonia detection from chest X-ray images, while Grad-CAM enhanced model interpretability by highlighting pathological regions [100].

A machine learning framework for osteoporosis prediction in patients with chronic kidney disease identified Artificial Neural Networks (ANN) as the best-performing model, achieving an AUC of 0.930 using eight key clinical features [101]. A disease prediction mobile application employing Principal Component Analysis (PCA) and machine learning classifiers identified Support Vector Machine (SVM) as the

best-performing model with an accuracy of 97.62% [102]. A comparative evaluation of KNN, Random Forest, SVM, and Linear Discriminant Analysis (LDA) for brain tumor classification from MRI images found that SVM achieved the highest accuracy of 90% [103].

A feature selection-based study for malignant disease prediction demonstrated that XGBoost achieved perfect recall (1.00), emphasizing the importance of optimizing evaluation metrics for clinical decision-making [104]. Several recent studies have further demonstrated the effectiveness of machine learning and deep learning techniques for disease diagnosis, prediction, and clinical decision support across diverse healthcare applications [105, 106, 107, 108, 109, 110, 111, 112, 113].

Rehman et al. [114] proposed a lightweight machine learning framework for pneumonia detection using 5,856 pediatric chest X-ray images. They applied Principal Component Analysis (PCA) for feature reduction and evaluated Support Vector Machine (SVM), K-Nearest Neighbors (KNN), and Random Forest (RF) classifiers. Among these, the SVM achieved the best performance with 94.97% accuracy, 97.54% sensitivity, 88.01% specificity, and an AUC-ROC of 0.98. The framework also addressed class imbalance and was validated using cross-validation and bootstrapping to ensure robustness. A lightweight machine learning framework for pneumonia detection using pediatric chest X-ray images achieved 94.97% accuracy, 97.54% sensitivity, 88.01% specificity, and an AUC-ROC of 0.98, with Support Vector Machine (SVM) outperforming KNN and Random Forest classifiers [114].

V. EXPLAINABLE AI AND INTERPRETABILITY METHODS

The **black-box** nature of advanced AI models like CNNs and Transformers hinders clinical adoption despite their high diagnostic accuracy. Explainable AI (XAI) addresses this by providing transparency and interpretability essential for clinical trust and regulatory compliance. As conceptually illustrated in the flowchart (Figure 2), **Input Data** is processed by the **ML Model**, where XAI tools like **SHAP Analysis** and **LIME Analysis** are explicitly integrated. These tools generate **Feature Influence Explanations** that, alongside the **Model Output**, contribute directly to two critical outcomes: **Increased Trustworthiness** and **Improved Interpretability**. XAI techniques operate at *global* (overall model behavior) and *local* (individual prediction) interpretability levels, with model-agnostic methods being particularly valuable for healthcare applications.

A. SHAP (SHapley Additive exPlanations)

SHAP is a unified framework based on cooperative game theory that assigns each feature an importance value for a particular prediction [118]. The Shapley value ϕ_i for feature i is calculated as:

$$\phi_i = \sum_{S \subseteq F \setminus \{i\}} \frac{|S|!(|F| - |S| - 1)!}{|F|!} [f(S \cup \{i\}) - f(S)] \quad (21)$$

TABLE III: Machine Learning and Deep Learning Applications in Healthcare (Part 1: Cardiovascular, Neurological, Oncology, Brain Tumor, and Pneumonia Applications)

Ref.	Focus Area	Model	Features/Approach	Dataset	Acc. (%)
Cardiovascular Applications					
[21]	Heart Disease	J48 Decision Tree	Clinical attributes with pruning	Clinical data	99.62
[22]	Cardiac Disease	SVM	Age, cholesterol, max heart rate	Clinical dataset	95.34
[23]	Heart Disease	LightGBM/XGBoost	Clinical + lifestyle (XAI)	300,000 patients	91.94
[24]	Coronary Artery	Random Forest	Bald Eagle Search optimization	Clinical data	92.00
[25]	Cardiovascular	Random Forest	SHAP + LIME interpretability	Clinical dataset	98.00
[26]	Heart Attack	KNN	Clinical parameters	Medical dataset	92.00
[?]	Heart Attack	Random Forest	Smart healthcare system	Medical dataset	99.89
[28]	Heart Failure	CatBoost	F1-score + ROC analysis	Clinical dataset	—
[29]	Heart Disease	XGBoost	Clinical + demographic	Kaggle dataset	93.00
[30]	Heart Disease	Gradient Boosting	12 ML models compared	Cleveland dataset	90.20
[31]	Heart Disease	Multiple classifiers	11 clinical parameters	Cleveland dataset	Varies
[32]	Heart Disease	SVM/Logistic Regression	WEKA vs Scikit-Learn	Clinical dataset	90.67 (WEKA)
[33]	CVD	CNN	ANOVA feature selection, 5-fold CV	Cleveland dataset (1,025)	98.54
Neurological Applications					
[34]	Parkinson's Epilepsy	Gradient Boosting/KNN	Clinical + EEG data	Medical dataset	89.00 / 85.00
[35]	Parkinson's Disease	Random Forest	SHAP interpretability	Clinical + demographic	93.00
[36]	Meningitis (Neonatal)	Logistic Regression	Risk factor analysis	535 neonates	AUC 0.765
[37]	Meningitis	LASSO Regression	6 critical CSF features	Clinical cohorts	AUC 0.921
[38]	Meningitis (Neonatal)	Logistic Regression	32 clinical variables	433 neonates	89.00
[115]	Meningitis (Preterm)	Logistic Regression	Clinical predictors	201 preterm infants	C-index 0.849
[?]	Brain Tumor	U-Net/ResNet	Multi-parametric MRI	Various	78.80–93.80
[40]	Brain Tumor	EfficientNet	CNN/Swin Transformer comparison	MRI multi-class	98.72
[41]	Brain Tumor	ResNet50	VGG16/U-Net comparison	Public MRI	95.35
[116]	Brain Tumor	SVM	Hybrid ML/DL review	Brain Tumor MRI	95.00
Oncology Applications					
[42]	Breast Cancer	Logistic Regression	11 clinical features	Clinical dataset	91.67
[43]	Breast Cancer	Random Forest	30 nuclear features	WDBC dataset	92.64
[44]	Breast Cancer	Random Forest	Clinical data	900 patients	96.00
[45]	Breast Cancer	Multimodal AI	Clinical + MG + ultrasound	5,000 patients	90.10
[46]	Breast Cancer	SVM	Clinical features	Wisconsin dataset	97.66
[47]	Breast Cancer	LightGBM	Feature selection	Biopsy data	90.74
[48]	Breast Cancer	SVM	Clinical features	Wisconsin dataset	High
[49]	Breast Cancer	Decision Tree	Preprocessing + selection	Clinical dataset	93.56
[50]	Breast Cancer	J48	Feature selection	Wisconsin dataset	93.56
[51]	Breast Cancer	Logistic Regression	Clinical features	Medical dataset	97.00
[52]	Breast Cancer	Random Forest	Transcriptomic data	Gene data	AUC 0.913
[53]	Breast Cancer	KNN	Clinical features	Medical dataset	98.40
[54]	Breast Cancer	Random Forest/SVM	Feature selection + CV	Wisconsin dataset	High
Pneumonia & Respiratory Applications					
[114]	Pneumonia	SVM	PCA, Cross-Validation, bootstrapping	Chest X-ray (5,856)	94.97
[100]	Pneumonia	VGG-16 + SVM	FAWT + Grad-CAM visualization	Chest X-ray	97.51

where F is the set of all features, S is a subset of features excluding i , and $f(S)$ is the model prediction using only feature subset S . The SHAP values satisfy the efficiency property:

$$\sum_{i=1}^M \phi_i = f(x) - E[f(X)] \quad (22)$$

where M is the number of features, $f(x)$ is the model prediction for instance x , and $E[f(X)]$ is the expected model prediction. In clinical applications, SHAP provides both local explanations (for individual patients) and global insights (by aggregating many local explanations). For breast cancer diagnosis using the Wisconsin Breast Cancer Dataset SHAP can identify the most critical biomarkers driving malignant

vs. benign classifications. As shown in Figure 3, features like worst radius, worst perimeter, and mean concave points emerge as the most important predictors. In breast cancer diagnosis, SHAP reveals that tumor morphological characteristics such as radius, texture, and perimeter measurements are key determinants of malignancy. The summary plot demonstrates how each feature affects the model output, with feature values color-coded to show their impact on prediction outcomes. This method offers theoretical guarantees from cooperative game theory and handles feature interactions effectively, though it can be computationally expensive for large datasets.

B. LIME (Local Interpretable Model-agnostic Explanations)

LIME explains individual predictions by approximating the complex black-box model with simple, interpretable model

TABLE IV: Machine Learning and Deep Learning Applications in Healthcare (Part 2: Ophthalmology, Dermatology, Nephrology, and Hepatology Applications)

Ref.	Focus Area	Model	Features/Approach	Dataset	Acc. (%)
Ophthalmology Applications					
[62]	Cataract	DCEN (Dual-stream)	ResNet-based grading	Fundus (1,340)	97.62
[63]	Cataract	CataractNet	CNN architecture	Multi-dataset	High
[64]	Eye Disease	YOLOv8n-cls	Fundus images	Multi-source (5,887)	94.00
[65]	Cataract	Mobile CNN	Smartphone-based	Mobile eye images	98.79
[66]	Cataract	Custom CNN	16-layer architecture	Clinical images	High
[67]	Cataract	BPNN	GLCM texture features	504 images	—
[68]	Cataract	SVM	Smartphone-based	Eye scans	95.00
[69]	Cataract	KNN	OpenCV processing	Mobile images	83.07
[70]	Cataract	SVM-RBF	Blood vessel features	Fundus images	95.20
[71]	Cataract	k-NN	GLCM texture analysis	Eye images (160)	97.50
Dermatology Applications					
[55]	Skin Lesion	HPDT	Prototypical networks	235,268 images	60.87
[117]	Skin Disease	Stacking Ensemble	ResNet50 + Logistic Regression	4,141 images	96.00
[56]	Skin Disease	Logistic Regression/SGD	Feature selection	Medical dataset	>98.00
[57]	Skin Cancer	Multiple DL	CNNs/Transformers/Ensembles	Various	>98.00
[58]	Skin Disease	Hybrid Ensemble (HEML)	Image preprocessing + 15 learners	Medical images	95.50
[59]	Skin Disease	Various ML	Image vs tissue-based	Survey data	50–100
[60]	Skin Disease	SVM	Dermoscopic features	ISIC/HAM10000	95.00–97.00
[61]	Skin Disease	SVM	Image features	Medical images	98.80
Nephrology Applications					
[82]	Kidney Stone	Hybrid CNN-ResNet	Feature extraction + filtering	Ultrasound/CT	90.90
[83]	Kidney Stone	Inception-V3	Transfer learning	Hospital CT (8,209)	98.52
[84]	Kidney Stone	Custom CNN	Smartphone microscope	37 stones (222 images)	88.00
[85]	Kidney Disease	Ensemble VGG-16	Fine-tuned final layers	Kaggle CT (2,000)	96.00
[86]	Kidney Disease	InceptionV3	Segmentation + HPT	Kaggle CT (12,446)	99.96
[87]	Kidney Stone	Logistic Regression	Clinical features	Medical dataset	78.00
[88]	Kidney Stone	Image Processing	Morphological analysis	Ultrasound	96.82
[89]	Kidney Stone	Logistic Regression	Urine analysis	79 samples	93.00
Hepatology Applications					
[42]	Liver Disease	CNN	Deep learning features	BUPA/ILPD	75.55 / 72.00
[72]	Liver Disease	SSAE	Unsupervised feature learning	Ultrasound	97.20
[73]	Liver Tumor	Custom DL	CT scan segmentation	CT images	98.59
[74]	Liver Disease	2S-NNet	Ultrasound features	Community US	AUC 0.90–0.93
[75]	Liver Disease	LightGBM	Clinical features	ILPD dataset	63.00
[76]	Liver Disease	Random Forest	Clinical parameters	32,000 records	97.30
[77]	Liver Disease	Logistic Regression	Basic clinical data	10,007 patients	72.80
[78]	Liver Disease	Ensemble ML	SVM/RF/DT/XGBoost	Medical dataset	95.24
[79]	Liver Disease	SVM	Transcriptomic data	TG-GATEs DB	AUC 0.82–0.98
[80]	Liver Disease	Hybrid Ensemble	LIME + SHAP XAI	Medical dataset	98.38
[81]	Liver Disease	Ensemble RF	SHAP + LIME validation	Medical dataset	99.92

in the local vicinity of the instance being explained. The method works by generating perturbed samples around the instance and training a local surrogate model that approximates the black-box model's behavior locally. The core optimization objective of LIME is formalized as:

$$\xi(x) =_{g \in G} \mathcal{L}(f, g, \pi_x) + \Omega(g) \quad (23)$$

where:

- 1) $\xi(x)$: Explanation for instance x .
- 2) f : Original complex model (black-box)
- 3) g : Interpretable surrogate model (e.g., linear model)
- 4) $\mathcal{L}$: Loss function measuring how well g approximates f in the locality defined by π_x
- 5) π_x : Proximity measure defining the local neighborhood around x

- 6) $\Omega(g)$: Complexity penalty for the explanation model g .

In our breast cancer diagnosis application using the Wisconsin Breast Cancer Dataset, LIME provides instance level explanations that identify which clinical features most influenced each individual prediction. For example, as shown in Figure 4, features such as worst area, worst concave points, and worst perimeter consistently emerge as the most influential factors across multiple patient cases. The model achieved 95.6% accuracy in classifying malignant vs. benign tumors, with LIME revealing that larger tumor area ($> 500 \text{ mm}^2$) and higher concavity typically pushed predictions toward malignancy, while smaller, smoother tumors supported benign classifications. The distribution of feature impacts (Figure 5) demonstrates that certain features, such as *worst area*, consistently correlate with malignancy (positive impact scores). This

TABLE V: Machine Learning and Deep Learning Applications in Healthcare (Part 3: Endocrinology, Metabolic Disorders, and Other Specialties)

Ref.	Focus Area	Model	Features/Approach	Dataset	Acc. (%)
Endocrinology & Metabolic Disorders					
[90]	Diabetes	KNN/Naïve Bayes	10-fold cross-validation	Pima Indians	Varies
[91]	Diabetes	Random Forest	Early detection	Medical dataset	87.00
[92]	Diabetes	Genetic J48	Feature optimization	Pima Indian	95.80
[93]	Diabetes	Multiple ML	Classification + ensemble	Pima Indian	Varies
[94]	Diabetes	ANN/RF/K-means	Association rule mining	Pima Indian	Varies
[95]	Diabetes	Multiple classifiers	Risk detection	Medical dataset	Varies
[96]	Diabetes	J48	Clinical features	768 patients	High
[97]	Diabetes	Decision Tree	Gini impurity optimization	Pima Indian	High
[90]	Diabetes Prediction	KNN/Naïve Bayes	Data preprocessing, 10-fold CV	Pima Indians	Varies
Other Medical Specialties					
[99]	Neurological	SVM/CNN (ResNet/VGG)	MRI-based diagnosis	Various	Up to 99.00
[101]	Osteoporosis	ANN	8 clinical features	6,614 patients	AUC 0.930
[102]	Multi-disease	SVM	PCA feature reduction	Kaggle dataset	97.62
[103]	Brain Tumor	SVM	Statistical features	MRI images	90.00
[104]	Malignant Cases	XGBoost	Recall-focused	Clinical dataset	Recall 1.00
[98]	Healthcare (Review)	CNNs/Transformers	Medical imaging, genomics	Electronic records	—
General Healthcare & Review Studies					
[98]	Healthcare AI Review	CNNs/Transformers	Medical image analysis	Various healthcare data	—
[99]	Neurological Review	SVM/CNN	MRI-based diagnosis	441 studies reviewed	Up to 99.00
[59]	Skin Disease Review	Various ML	Image vs tissue-based methods	Survey data	50–100

Note: Abbreviations: SVM = Support Vector Machine, LightGBM = Light Gradient Boosting Machine, XGBoost = Extreme Gradient Boosting, KNN = K-Nearest Neighbors, LR = Logistic Regression, LASSO = Least Absolute Shrinkage and Selection Operator, DL = Deep Learning, CNN = Convolutional Neural Network, ResNet = Residual Network, VGG = Visual Geometry Group, YOLO = You Only Look Once, U-Net = U-shaped Network, DCEN = Dual-stream Cataract Evaluation Network, HPDT = Hierarchical Prototypical Decision Tree, SSAE = Stacked Sparse Auto-Encoder, RF = Random Forest, BPNN = Backpropagation Neural Network, GLCM = Gray Level Co-occurrence Matrix, AUC = Area Under the ROC Curve, ROC = Receiver Operating Characteristic, XAI = Explainable Artificial Intelligence, SHAP = SHapley Additive exPlanations, LIME = Local Interpretable Model-agnostic Explanations, FAWT = Flexible Analytic Wavelet Transform, PCA = Principal Component Analysis, ANN = Artificial Neural Network, CV = Cross-Validation, HEML = Hybrid Ensemble Machine Learning, MG = Mammography, CVD = Cardiovascular disease, HPT = Hyperparameter Tuning

local interpretability allows clinicians to verify that the model relies on medically relevant features, building trust in the diagnostic system. While LIME is intuitive and computationally efficient, its explanations can be sensitive to the perturbation process. Furthermore, it only provides local insights without offering a global perspective on model behavior. Nevertheless, in clinical applications like cancer diagnosis, LIME’s case specific, human understandable explanations are invaluable for validation and decision support [119].

C. Grad-CAM (Gradient-weighted Class Activation Mapping)

It is a technique for CNNs, especially useful in medical image analysis [120]. It generates visual explanations as heatmaps that highlight image regions contributing most to the model’s decision, based on gradients flowing into the final convolutional layer. As shown in brain tumor MRI studies [121, 122] and your synthetic breast cancer simulation, Grad-CAM effectively pinpoints tumor or lesion areas, confirming the model’s focus on clinically relevant regions. The method computes a weighted sum of the final convolutional feature maps ($\mathbf{A}^k$), where the weights (α_k^c) are obtained from the spatial average of the gradients ($\frac{\partial y^c}{\partial \mathbf{A}_{ij}^k}$) for the target class (c). The two defining equations are the Neuron Importance Weight (α_k^c) and the Grad-CAM Heatmap ($\mathbf{L}_{\text{Grad-CAM}}^c$):

- 1) Neuron Importance Weight:

$$\alpha_k^c = \frac{1}{Z} \sum_i \sum_j \frac{\partial y^c}{\partial \mathbf{A}_{ij}^k}$$

- 2) Grad-CAM Heatmap:

$$\mathbf{L}_{\text{Grad-CAM}}^c = \text{ReLU} \left(\sum_k \alpha_k^c \mathbf{A}^k \right)$$

The Grad-CAM process begins by calculating the Neuron Importance Weight (α_k^c), derived from the gradient ($\frac{\partial y^c}{\partial \mathbf{A}_{ij}^k}$) using Global Average Pooling (GAP) to quantify each feature map’s relevance to the classification score y^c . These weights are linearly combined with the corresponding feature maps $\mathbf{A}^k$ to generate the final Grad-CAM Heatmap ($\mathbf{L}_{\text{Grad-CAM}}^c$), followed by Rectified Linear Unit (ReLU) to retain only positive activations. The resulting visualizations, shown in Fig. 6 and Fig. 7, highlight tumor-representative regions that influenced the malignant prediction, while Table ?? provides corresponding model interpretation details.

D. Model-Specific and Intrinsic Interpretability Methods

Model-specific interpretability techniques leverage internal model architectures to generate faithful explanations. **Attention mechanisms** in Transformers provide natural interpretability through attention weights that highlight relevant

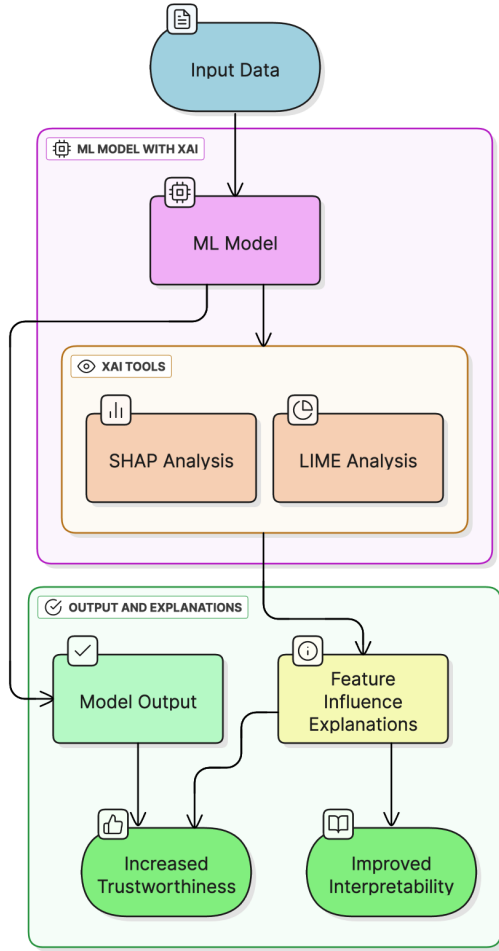


Fig. 2: Explainable AI (XAI) workflow. SHAP and LIME provide feature explanations to improve model trustworthiness and interpretability.

input regions [99]. For CNNs, **saliency maps** and **guided backpropagation** visualize influential pixels by computing output gradients with respect to inputs:

$$M(x) = \left\| \frac{\partial f_c(x)}{\partial x} \right\| \quad (24)$$

Tree-based models offer inherent interpretability through **rule extraction**, converting decision paths into human-readable "if-then" rules that align with clinical guidelines [30]. **Inherently interpretable models** like Generalized Additive Models (GAMs) provide transparency by design:

$$g(E[y]) = \beta_0 + f_1(x_1) + f_2(x_2) + \dots + f_p(x_p) \quad (25)$$

These methods deliver architecture-specific insights crucial for clinical validation of AI reasoning processes.

E. Comparative Analysis of XAI Techniques

Each XAI method offers distinct advantages and limitations in healthcare applications, as summarized in Table VI. SHAP provides the most mathematically rigorous explanations but

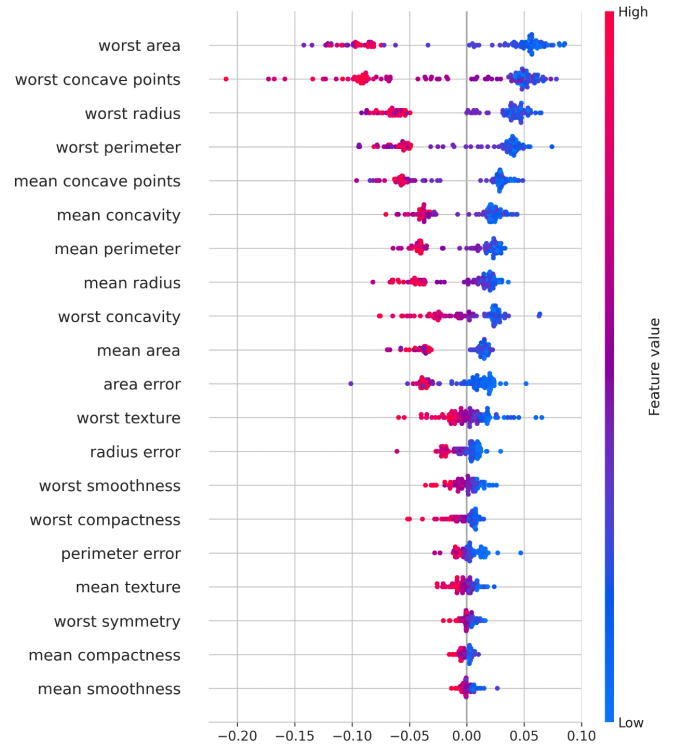


Fig. 3: SHAP summary plot showing feature importance and impact direction for breast cancer classification. Red features push predictions toward malignancy, while blue indicates benign influence.

requires significant computational resources. LIME offers intuitive local explanations but may lack consistency across similar cases. Grad-CAM delivers excellent visual interpretability for imaging data but is limited to CNN architectures. The choice of XAI technique often depends on the specific clinical context, with SHAP being preferred for tabular clinical data, LIME for mixed data types, and Grad-CAM for medical imaging applications.

TABLE VI: Comparison of XAI Techniques in Healthcare

Technique	Key Characteristics
SHAP	Level: Global & Local Advantages: Theoretical guarantees, consistent explanations, handles feature interactions Limitations: Computationally expensive, complex for non-technical staff
LIME	Level: Local Advantages: Intuitive, fast computation, model-agnostic, immediate insights Limitations: Explanation instability, sensitive to parameters, no global view
Grad-CAM	Level: Local (Visual) Advantages: Visual heatmaps, no retraining, aligns with radiological practice Limitations: Limited to images/CNNs, may miss fine details

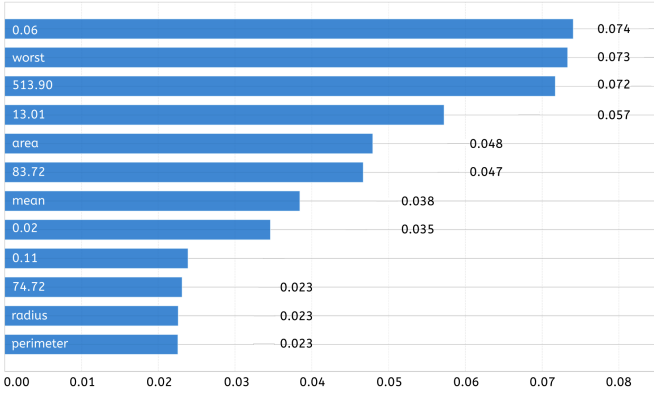


Fig. 4: Aggregate feature importance across 50 LIME explanations. The chart displays the Mean Absolute Impact Score (Mean ILIME Feature Importance) for each clinical parameter, with the features ordered by their importance, from the most influential (top bar) to the least influential (bottom bars) in breast cancer diagnosis.

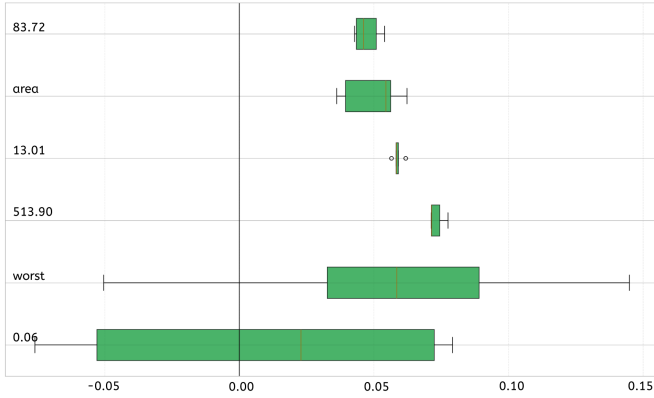


Fig. 5: Distribution of feature impact scores derived from LIME. Scores to the right (positive) support Malignant scores to the left (negative) support Benign.

VI. A DECISION FRAMEWORK FOR DOMAIN-SPECIFIC XAI SELECTION

The selection of appropriate Explainable AI (XAI) techniques in healthcare is not a one-size-fits-all decision. The optimal method depends on multiple factors, including data modality, model architecture, clinical context, and stakeholder requirements. This section provides a comprehensive framework for domain-specific XAI selection, synthesizing findings from recent systematic reviews and empirical studies to guide researchers and clinicians in choosing the most suitable interpretability approach for their specific healthcare application. Recent systematic reviews have established the prevalence of different XAI techniques across healthcare domains. A comprehensive analysis of 70 studies revealed that SHAP is the most frequently applied technique (54% of studies), followed by LIME (30%), Grad-CAM (23%), and attention mechanisms (20%). The same review found that deep learning models, particularly CNNs, RNNs, LSTMs, and Transformers,

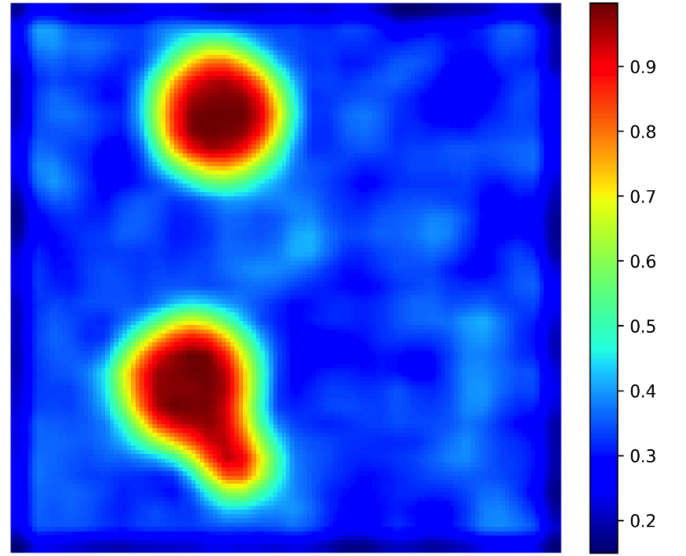


Fig. 6: Raw Grad-CAM Heatmap: The raw output of $L_{\text{Grad-CAM}}^c$, resized to the original image dimensions, showing relative activation intensity across the image.

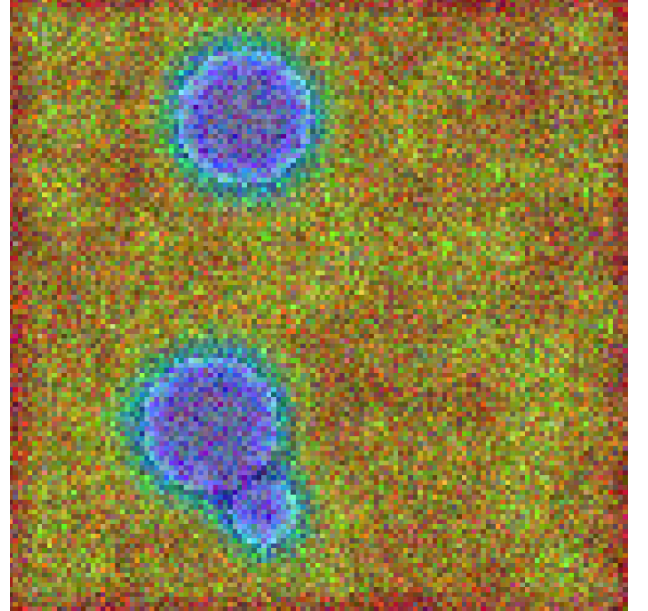


Fig. 7: Interpretable Result (Overlay): The heatmap superimposed on the original image, visually confirming that the model focused on the irregular pattern (tumor surrogate) to predict malignant.

account for 76% of healthcare AI applications, while tree-based models represent 24% of implementations. However, the effectiveness of these techniques varies considerably across data modalities and clinical tasks.

A. Framework Dimensions

The selection of XAI techniques should consider four primary dimensions: data modality, model architecture, ex-

planation scope, and clinical context. Table VII provides a comprehensive mapping of XAI techniques to specific healthcare domains and use cases, while Table VIII presents the underlying reasons and clinical justifications for these recommendations.

VII. PERSONALIZED TREATMENT AND CLINICAL APPLICATIONS

A. AI-Driven Personalized Medicine

Artificial Intelligence (AI) is reshaping personalized medicine by enabling tailored treatment strategies based on individual patient characteristics, genetic profiles, and clinical history. Traditional “one-size-fits-all” medical approaches are gradually being replaced by precision medicine frameworks that leverage machine learning to analyze complex, multi-modal data—including genomics, proteomics, metabolomics, and clinical parameters—as summarized in Table IX.

Genomic-Based Personalization: AI algorithms play a pivotal role in identifying biomarkers and genetic signatures that predict how patients will respond to specific treatments. Deep learning models can process large-scale genomic data to detect rare variants that influence drug metabolism and therapeutic efficacy. In oncology, convolutional neural networks (CNNs) analyze genomic and transcriptomic profiles to predict chemotherapy sensitivity and guide the selection of targeted therapies based on tumor-specific mutations. Reinforcement learning approaches further enhance personalization by optimizing dosing regimens continuously adapting to a patient’s treatment response and side effect patterns, as detailed in Table IX.

Multi Omics Integration: The integration of diverse biological datasets through AI enables a more holistic understanding of disease mechanisms and patient subtypes. Graph neural networks (GNNs) model intricate relationships between genomic variations, gene expression, and protein interactions to uncover new therapeutic targets. Autoencoder-based architectures learn compact, informative representations of high-dimensional omics data, helping to identify patient subgroups with distinct molecular signatures and treatment needs. These integrative approaches have demonstrated improved predictive accuracy for drug response compared to models relying on single data types, highlighting the advantages summarized in Table IX.

B. Clinical Decision Support Systems

AI-powered **Clinical Decision Support Systems (CDSS)** are transforming the way clinicians make decisions by providing data-driven, evidence-based insights directly at the point of care. These systems combine patient-specific data with vast medical knowledge to assist in diagnosis, treatment planning, and risk assessment, as outlined in Table IX.

Diagnostic Decision Support: Advanced CDSS employ deep learning models to analyze a combination of medical images, laboratory results, and clinical notes

for comprehensive diagnostic support. Transformer-based models are capable of processing multimodal patient data to produce ranked differential diagnoses with associated confidence scores. Furthermore, knowledge graphs integrate structured medical ontologies with real-world clinical evidence, generating contextualized recommendations that account for comorbidities, drug interactions, and established medical guidelines. In several domains—such as radiology, pathology, and emergency medicine—these systems have demonstrated diagnostic performance on par with human specialists, reinforcing the clinical potential identified in Table IX.

Risk Stratification and Early Warning: Machine learning-based risk prediction models help identify patients who are most likely to experience adverse clinical events, allowing for timely intervention. Gradient boosting algorithms analyze EHR data to predict outcomes such as hospital readmissions, sepsis onset, and patient deterioration. Real-time monitoring systems interpret streaming physiological data from bedside sensors to detect subtle changes preceding critical events. In parallel, natural language processing (NLP) techniques extract valuable clinical insights from unstructured physician notes, identifying patients who may benefit from additional testing or specialist referrals—addressing the real-world implementation challenges discussed in Table IX.

Treatment Pathway Optimization: Reinforcement learning frameworks are increasingly used to optimize complex treatment pathways by learning from historical patient outcomes. These systems model treatment selection as a sequential decision-making problem—balancing short-term improvements against long-term health outcomes. Multi-armed bandit algorithms support adaptive clinical trial designs by dynamically allocating treatment arms, expediting the discovery of effective therapies while maintaining ethical integrity. Additionally, Markov decision processes (MDPs) encode clinical guidelines into actionable policies that can be personalized for each patient based on their unique characteristics and preferences, as reflected in the AI technologies column of Table IX.

VIII. FUTURE TRENDS

A. Mobile Health (mHealth) Applications

The combination of artificial intelligence with mobile health technologies is a game-changer in healthcare delivery, especially for making medical services more available in areas that don’t have enough of them. Mobile health apps use the fact that smartphones and wearable devices are everywhere to allow for constant health monitoring, remote diagnosis, and personalized interventions. These deployable solutions show how AI algorithms may be used in real-world healthcare settings, filling the gap between research and clinical use.

Mobile Diagnostic Applications: Numerous studies have shown that it is possible to use AI models on mobile devices

TABLE VII: Comprehensive Framework for Domain-Specific XAI Selection in Healthcare

Healthcare Domain	Data Modality	Recommended XAI	Primary Rationale	Key Considerations
Oncology (Tumor Detection)	Medical Imaging (MRI, CT, X-ray)	Grad-CAM	Provides visual heatmaps aligning with radiological workflow; highlights lesion boundaries and suspicious regions	CNNs/Transformers; combine with SHAP for multimodal data
Oncology (Prognosis)	Tabular + Genomic	SHAP + LIME	Quantifies feature contributions; handles interactions in multi-omic data; provides consistent global interpretations	Tree-based models (RF, XGBoost); SHAP preferred for global consistency
Cardiology (Arrhythmia)	ECG/EEG Signals	SHAP + Attention	Temporal feature attribution; attention highlights relevant signal segments	LSTM/CNN architectures; SHAP for feature importance across time
Cardiology (Risk Prediction)	Tabular EHR Data	SHAP (Global) + LIME (Local)	Global feature importance for risk factors; local explanations for individual patient risk assessment	Tree-based models favored; SHAP computationally efficient for RF/GB
Neurology (Brain MRI)	MRI/CT Imaging	Grad-CAM + SHAP	Visual localization of pathology; SHAP for clinical/demographic features	CNN/U-Net architectures; modality-specific performance
Neurology (Alzheimer's)	Multimodal (MRI + Tabular + Genetic)	SHAP + LRP	Multi-source integration; LRP for layer-wise MRI interpretation; SHAP for tabular features	GNN + CNN; dashboard visualization for clinicians
Dermatology	Dermoscopic Images	Grad-CAM + SHAP	Visual verification of lesion features; highlights morphological characteristics	ViT/CNN architectures; combined with LIME for instance-level insights
Clinical Decision Support	Tabular EHR Data	SHAP (Global) + LIME (Local) Balance of model understanding and case-specific explanations	Tree-based models favored; SHAP computationally efficient for RF/GB	
Medical Text/NLP	Clinical Notes	LIME + SHAP	Token-level attribution for sentiment/risk identification; local explanations for clinical validation	LLM/Transformer architectures; SHAP for global patterns
Ultrasound Imaging	Ultrasound	LIME	Superior performance identified for US modality; preserves original model confidence	CNNs; clinician preference strongest for LIME
Hematology	Blood Smear Images	Grad-CAM + LIME + SHAP	Comprehensive morphological feature visualization; multi-method validation	Lightweight CNN architectures; visual explanations critical for rare cell types
Multimodal Integration	Images + Tabular + Text	Grad-CAM + SHAP	Modality-specific explanations; comprehensive decision understanding	Late/intermediate fusion architectures; combined interpretability layers

TABLE VIII: Clinical Rationale and Evidence-Based Justification for XAI Selection in Healthcare Domains

Healthcare Domain	Recommended XAI	Clinical Rationale	Supporting Evidence
Oncology (Tumor Detection)	Grad-CAM	Visual explanations mimic radiological interpretation; heatmaps highlight suspicious regions for biopsy guidance	98.72% accuracy in brain tumor classification; heatmaps correlate with radiologist annotations
Oncology (Prognosis)	SHAP + LIME	Identifies key biomarkers; multi-omic feature importance; survival prediction transparency	SHAP reveals top 5-10 genomic features driving metastasis risk; supports treatment planning
Cardiology (Arrhythmia)	SHAP + Attention	Temporal pattern identification; highlights critical ECG segments; clinical feature attribution	Attention weights correlate with P-wave, QRS complex, T-wave abnormalities; SHAP identifies age/cholesterol impact
Cardiology (Risk Prediction)	SHAP + LIME	Global risk factor identification; individual risk communication; supports preventive interventions	SHAP consistently identifies age, cholesterol, blood pressure as top predictors; LIME enables patient-specific counseling
Neurology (Brain MRI)	Grad-CAM + SHAP	Visual confirmation of pathology localization; demographic/clinical context integration	Grad-CAM localizes atrophy/tumor regions; SHAP identifies age, gender, cognitive score contributions
Neurology (Alzheimer's)	SHAP + LRP	Multi-modal integration; layer-wise interpretation for early detection; biomarker identification	LRP highlights hippocampal atrophy; SHAP identifies genetic (APOE) and cognitive feature importance
Dermatology	Grad-CAM + SHAP	ABCD rule alignment; visual lesion morphology verification; morphological feature attribution	Grad-CAM highlights asymmetry, border, color, diameter features; SHAP quantifies clinical feature contributions
Clinical Decision Support	SHAP + LIME	Global guideline alignment; case-specific reasoning; clinician-AI collaborative decision-making	SHAP ensures model reasoning follows medical consensus; LIME provides case-specific validation
Medical Text/NLP	LIME + SHAP	Patient-friendly explanations; clinical note risk identification; medical concept extraction	LIME highlights key symptoms, medications; SHAP identifies global linguistic patterns
Ultrasound Imaging	LIME	Superior performance for US; preserves confidence; intuitive for clinicians	LIME outperforms Grad-CAM for US; higher clinician acceptance in studies
Hematology	Grad-CAM + LIME + SHAP	Rare cell classification; morphological feature validation; multi-method verification	Grad-CAM visualizes cell morphology; LIME/SHAP provide complementary verification for rare diseases
Multimodal Integration	Grad-CAM + SHAP	Comprehensive patient understanding; modality-specific explanation; trust enhancement	Grad-CAM for imaging; SHAP for clinical data; integrated dashboards improve clinician confidence

TABLE IX: Comprehensive Overview of AI Applications in Personalized Treatment and Clinical Decision Support

Application Area	AI Technology	Key Capabilities	Clinical Benefits	Implementation Challenges
Personalized Medicine	Graph Neural Networks Autoencoders Reinforcement Learning	Multi-omics data integration Biomarker discovery Individualized dosing optimization	Improved treatment efficacy Reduced adverse events Targeted therapy selection	Data heterogeneity Interpretability requirements Regulatory validation
Clinical Decision Support	Transformer Architectures Knowledge Graphs Gradient Boosting Machines	Multimodal data analysis Evidence-based recommendations Real-time risk stratification	Enhanced diagnostic accuracy Proactive intervention Reduced medical errors	Workflow integration Alert fatigue management Clinical acceptance
Treatment Recommendation	Multi-objective Reinforcement Learning Bayesian Neural Networks Federated Learning	Balanced treatment optimization Uncertainty quantification Continuous learning from real-world data	Personalized therapy sequences Informed decision making Adaptive treatment strategies	Causal inference limitations Data privacy concerns Validation complexity
Drug Discovery & Development	Generative AI Molecular Dynamics Simulation Predictive Toxicology Models	De novo drug design Binding affinity prediction Toxicity and efficacy forecasting	Accelerated discovery timeline Reduced development costs Improved success rates	Experimental validation Regulatory acceptance Computational complexity
Chronic Disease Management	Temporal Convolutional Networks LSTM/GRU Architectures Digital Twin Modeling	Disease progression prediction Treatment response monitoring Personalized intervention timing	Preventive care optimization Reduced complications Improved quality of life	Long-term data requirements Model drift over time Patient engagement

to treat a wide range of medical ailments. Rehman et al. [?] created the Cortex Vision app, which is a good example of this trend. It uses an SVM model with 95% accuracy to find cataracts in ocular photos taken with smartphone cameras. In the same way, Rajhan et al. [65] made a mobile app for detecting cataracts using CNN architecture that was 98.79% accurate. Agarwal et al. [69] made an Android-based system for detecting cataracts using KNN algorithms that was 83.07% accurate. These mobile solutions help people in resource-limited areas get the healthcare they need when specialists aren't available.

Wearable Health Monitoring: AI-powered wearable gadgets can do more than just diagnostics; they can also monitor your body constantly and provide you alerts when something is wrong. Smartwatches and fitness trackers that use machine learning can find arrhythmias, keep an eye on blood sugar levels, and keep track of how well people are taking their medications. These systems use lightweight deep learning models that are tailored for edge computing. They process sensor data locally to protect privacy and give real-time health information. Federated learning and IoT devices coming together make it possible to create tailored health models without having to collect data in one place. This solves privacy issues while keeping the models accurate.

IX. CONCLUSION

A. Key Findings and Insights

This in-depth study has explored the transformative impact of artificial intelligence in the healthcare landscape, revealing several key insights into the current state and future trajectory of medical AI. The evidence demonstrates that in some areas AI systems are consistently as good or better at conducting diagnoses as human experts. For instance, numerous studies indicate that the AI systems can detect brain tumors, cardiovascular risks, and medical images with 95% to 99% accuracy or more, as outlined in Table III, IV and V. Together, these technological advances highlight the potential of AI to significantly enhance diagnostic accuracy and efficiency in various healthcare fields, as summarized in Table X. With the growing integration of Explainable AI methodologies, clinical trust and adoption are key elements to consider. Techniques like SHAP, LIME, and Grad-CAM make AI judgments clear enough for healthcare workers to comprehend and check them. This is the solution to the so-called "black box" issue which has constrained AI use in clinical environments. The comparative analysis in Table VI shows how multiple XAI methods can be used together to improve diverse healthcare applications, from tabular clinical data to medical imaging. The information in Table X shows that explainability and trust are essential for successful clinical implementation. This translates to a different approach for various stakeholder groups and clinical settings.

In the present times, personalized treatment is one of the most significant advancements in medical care. AI makes precision medicine possible by taking into account a person's genetic profile, lifestyle, and medical history. The schemes presented in Table IX illustrate the possibility of making treatment plans unique to each patient rather than the tra-

TABLE X: Comprehensive Summary of Key Findings, Insights, and Recommendations for Healthcare AI Implementation

Category	Key Findings	Critical Insights	Strategic Recommendations
Technical Performance	AI models achieve 95–99% accuracy in diagnostics; deep learning excels in imaging; ensemble methods enhance robustness.	Performance varies across populations and settings; real-world accuracy often lower; computational costs remain high.	Adopt standardized evaluation frameworks, ensure external validation, and maintain continuous performance monitoring.
Explainability and Trust	XAI improves interpretability using SHAP, LIME, and Grad-CAM; interpretability increases clinician confidence.	Different XAI methods suit specific data types; explanations must align with clinical logic; balance accuracy and transparency.	Mandate explainability in clinical systems, tailor explanations to user roles, and validate their clinical relevance.
Clinical Integration	Workflow integration remains difficult; human–AI collaboration and mobile health solutions are emerging.	Seamless integration and clinician training are critical; organizational readiness varies widely.	Apply user-centered design, implement training initiatives, and develop change management strategies.
Ethical Considerations	Algorithmic bias and privacy concerns persist; accountability mechanisms are still limited.	Bias mitigation must be proactive; privacy-preserving approaches are advancing; legal liability remains unclear.	Establish bias auditing frameworks, promote federated learning, and define accountability standards.
Regulatory Compliance	Regulatory evolution is slow; global alignment limited; validation requirements increasing.	Traditional frameworks struggle with adaptive AI; pre-market approvals are lengthy; post-market surveillance essential.	Develop adaptive regulatory pathways, align international standards, and enforce ongoing monitoring.
Future Directions	Federated learning enhances privacy; generative AI improves efficiency; quantum computing offers new potential.	Technology progresses faster than adoption; infrastructure and workforce adaptation remain major barriers.	Invest in emerging technologies, strengthen digital infrastructure, and promote AI literacy and skill development.

ditional "one-size-fits-all" approach, through multi-omics integration, longitudinal data analysis and adaptive learning systems. As evidenced in Table X, there is significant potential in these personalised approaches in the treatment of chronic diseases and enhancing the effectiveness of complex therapies. However, they also require a robust data infrastructure and significant computing power to function effectively. Despite these technological advances, there are many ethical, legal and practical implementation issues. Algorithmic bias, data privacy issues, and problems with interoperability are still making it hard for AI to be widely used. At the same time, regulatory systems often have a hard time keeping up with how quickly AI technologies are changing. Table ?? presents the barriers to adoption of AI in healthcare environments, highlighting the complexity of applying AI in such a context. These are technological, organisational and social issues as demonstrated once more in Table X. Researchers, clinicians, policy makers and technology developers should collaborate to ensure safe and equitable implementation.

B. Recommendations for Clinical Implementation

Based on the comprehensive analysis presented in this review, we propose several evidence-based recommendations to facilitate the successful integration of AI technologies in healthcare settings. These recommendations span technical, ethical, and operational aspects of AI integration, offering a blueprint for healthcare stakeholders, technology developers, regulators, and policy makers. The strategic recommendations presented in Table X offer a roadmap for achieving success in the healthcare sector by tackling the challenges and leveraging

the opportunities afforded by AI.

Develop Standardized Evaluation Frameworks: Set up thorough validation protocols that look at AI systems from many angles, such as how well they work technically, how useful they are in the clinic, and how well they work in the real world. To ensure studies are assessed uniformly and to allow meaningful comparisons of the various AI approaches, these frameworks must be comprehensive enough to include the various metrics listed in Table II. Implementation should include rigorous external validation with independent data with diverse patient populations and clinical settings. As shown in Table X, ongoing performance monitoring is necessary to keep the model accurate and useful as clinical practices and patient populations change.

Prioritize Explainability and Interpretability: Insist on XAI techniques for all clinical AI systems; make sure explanations are suitable for all relevant stakeholders. It is important for the clinicians to have clear and intuitive case-specific explanations for AI recommendations. Patients should be made aware of the use of their information and what is being concluded. Regulatory approvals should include requirements on transparency and interpretability, along with validation to make sure that explanations are consistent with the clinical knowledge and reasoning processes, as shown in Table VI. The suggestions in Table X show how important it is to customize explanations for different user roles and make sure they are clinically relevant so that they help rather than make clinical decision-making harder.

Address Algorithmic Bias Proactively: Apply full bias detection and mitigation strategies throughout the entire process of AI development. This involves curating a variety of datasets, continuous monitoring for performance gaps within demographic groups and developing fairness-aware algorithms. Bias auditing procedures and equity impact assessments should be adopted by healthcare organizations to ensure that all the AI systems deployed reduce or do not worsen the existing health disparities. As summarized in Table X, mitigating biases requires proactive measures throughout the AI lifecycle, from data collection to deployment and monitoring.

Following the suggestions in Table X, healthcare will be able to responsibly and effectively use AI technologies. As stakeholders begin to proactively tackle the issues of technical, ethical, and operational hurdles, they can maximize the transformative power of AI, thereby improving patient outcomes, enhancing clinical efficiency, and expanding access to high-quality care globally. To turn black-box algorithms into clear and reliable clinical tools, all the stakeholders in various fields and sectors must collaborate, and do so over a long period of time, with the common purpose of ensuring that the welfare of patients is at the core of technological progress. The framework in Table X gives a complete guide to how to get through this complicated landscape, balancing innovation with responsibility to make sure that AI technologies help patients and healthcare systems around the world.

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